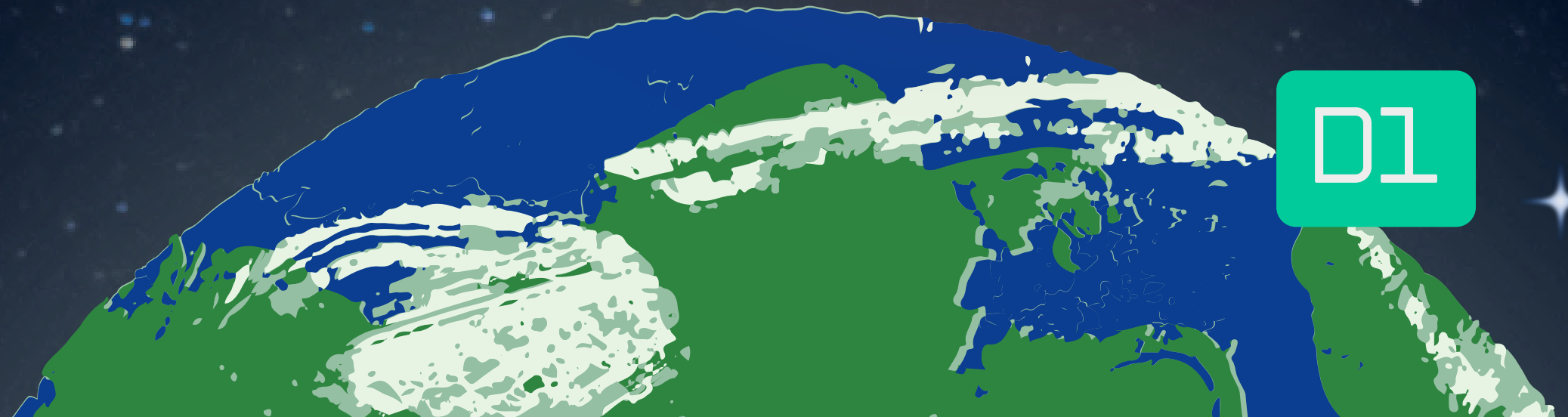
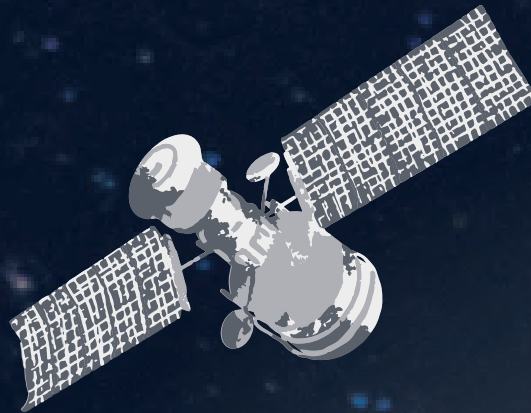




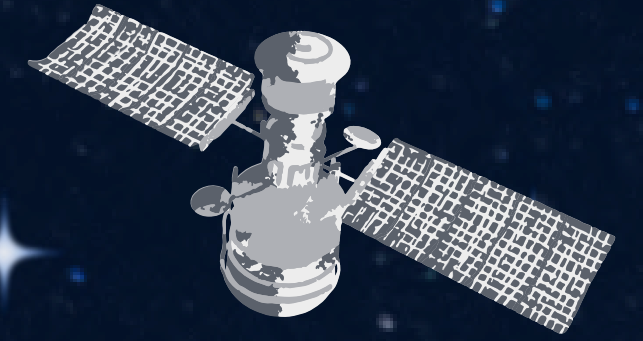
Facultad de Ingeniería
L. Guarino – J. Sorondo – A. Ferreira
Prof. Ignacio Boero

Procesamiento de imágenes **satelitales**



D1

Objetivos



¹ Satélites
NOAA

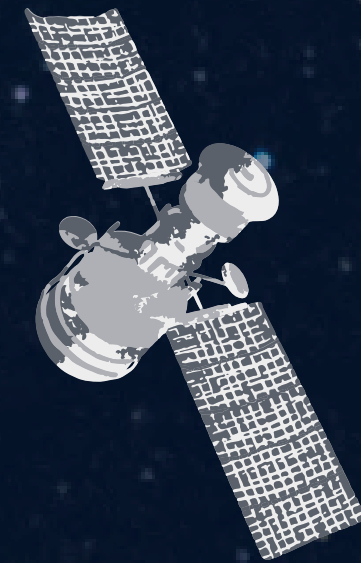
² Antena y
Grabaciones

³ Procesamiento
de la **Señal**

⁴ Nuestras
Imágenes

Importancia de los **Satélites**

Meteorología
Comunicaciones
Navegación
Investigación
Seguridad

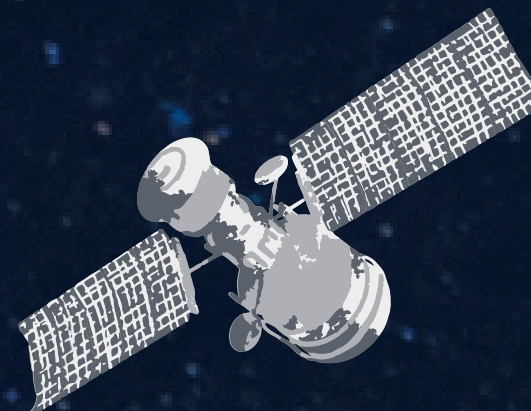
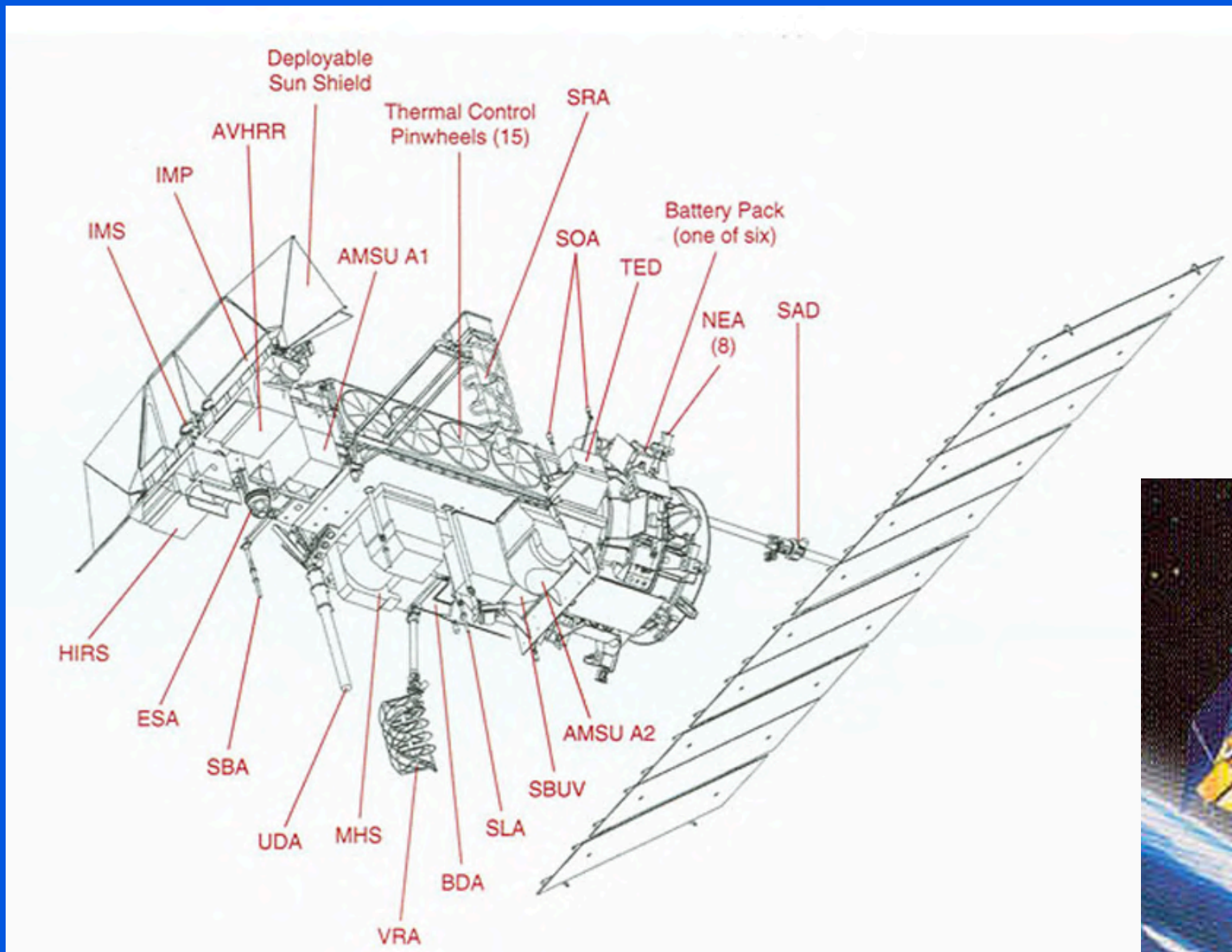


Sobre la NOAA

Administración Oceánica y Atmosférica
de los Estados Unidos de América



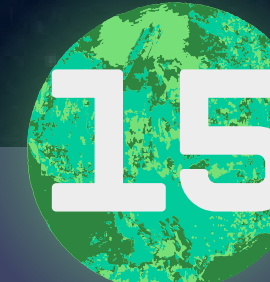
Tipos de satélites



137,1
MHZ



137,912
MHZ



137,62
MHZ

Sobre AVHRR

Radiómetro
Avanzado de muy alta resolución



**X15 revoluciones
diarias**

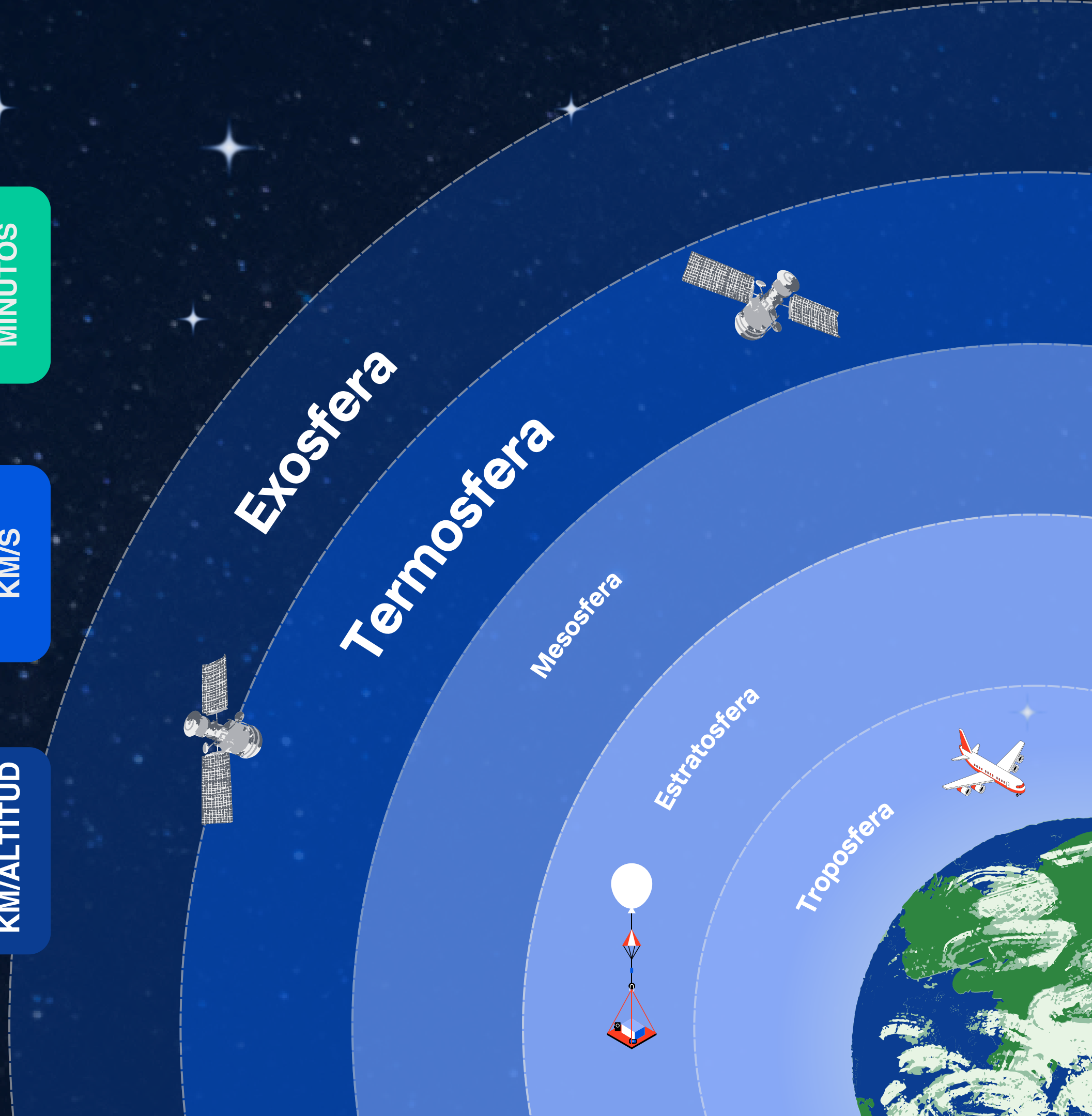
100 MINUTOS

**Tienen una velocidad
entre 7 y 10 KM/S**

10 KM/S

**Mantienen una altitud
de entre los 833 y
870 kilómetros**

870 KM/ALTITUD



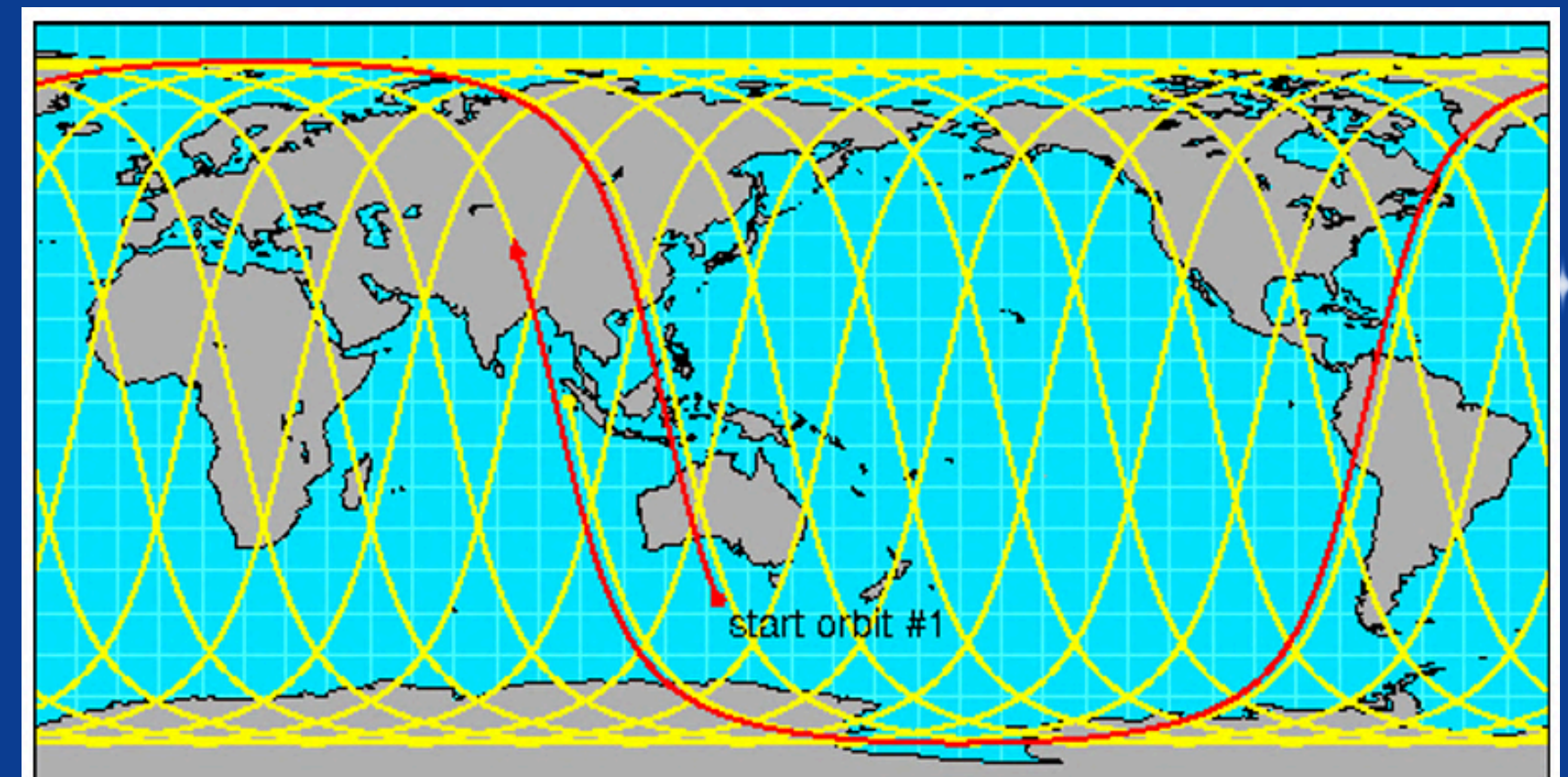
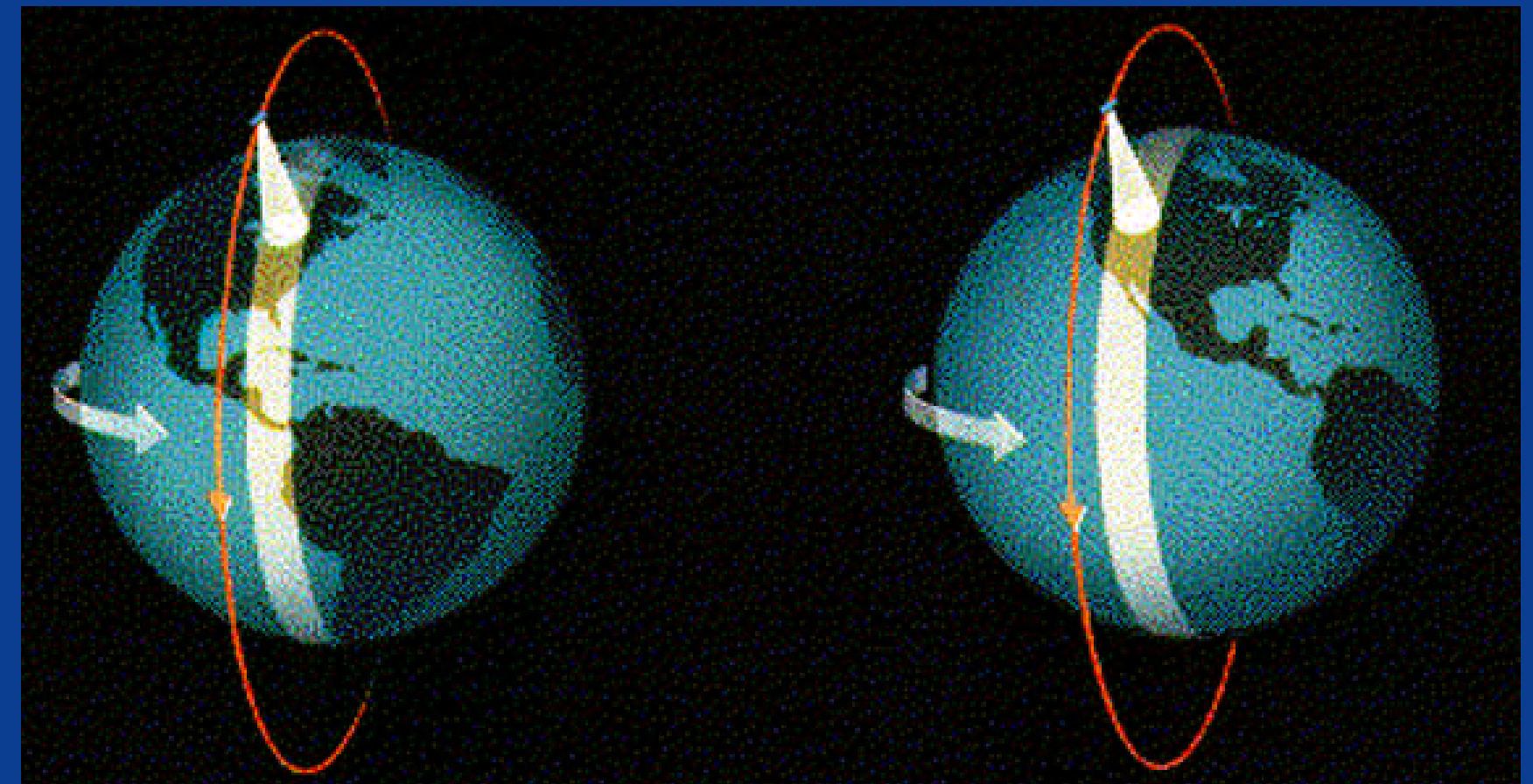
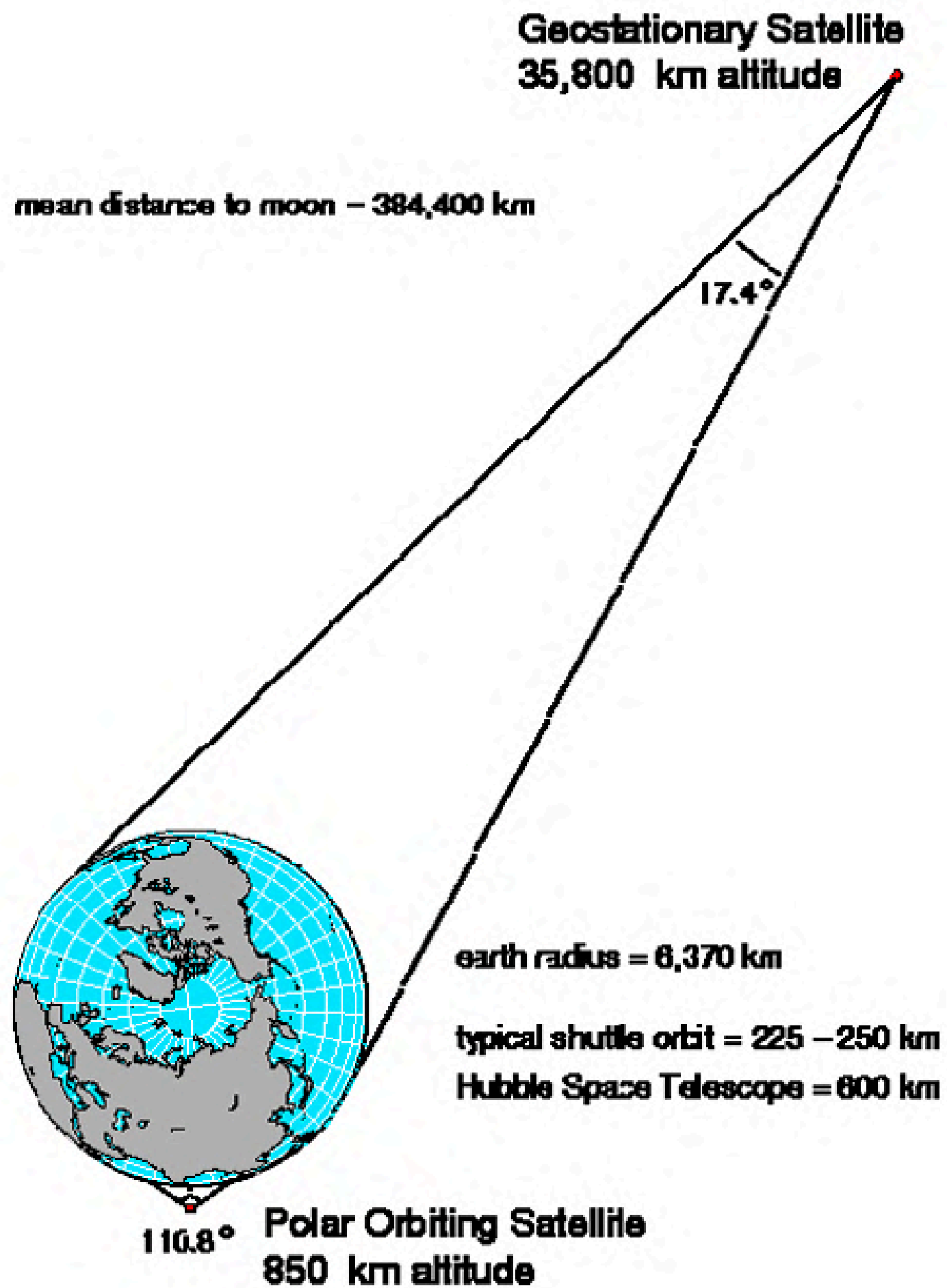
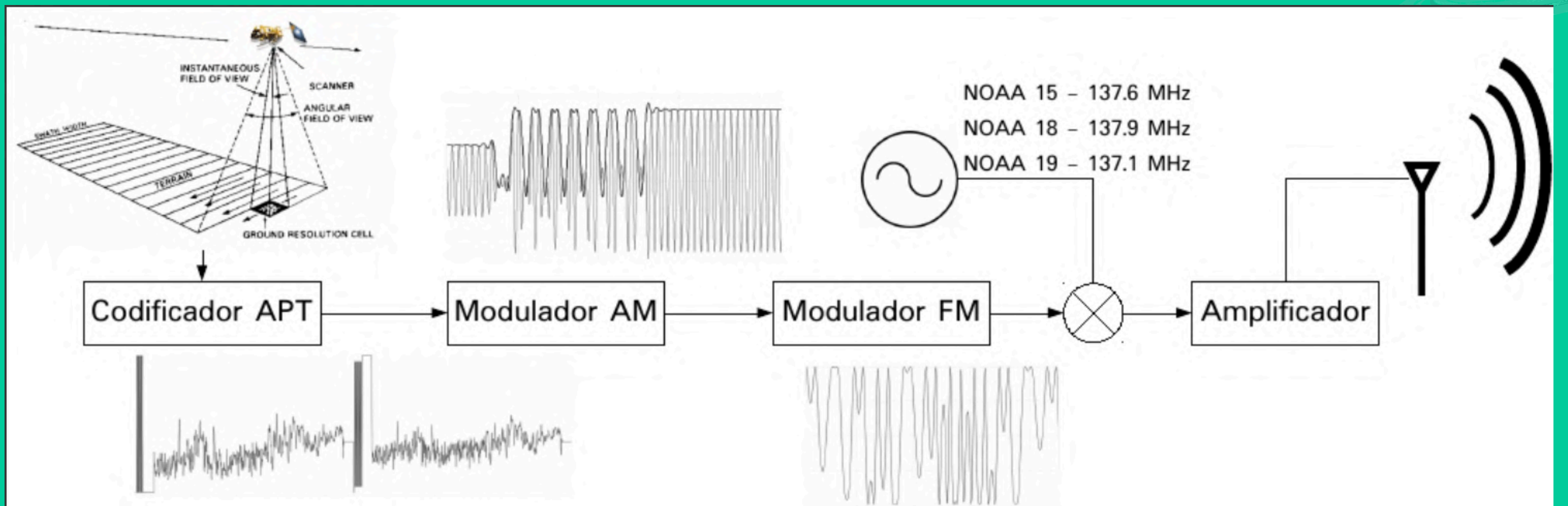


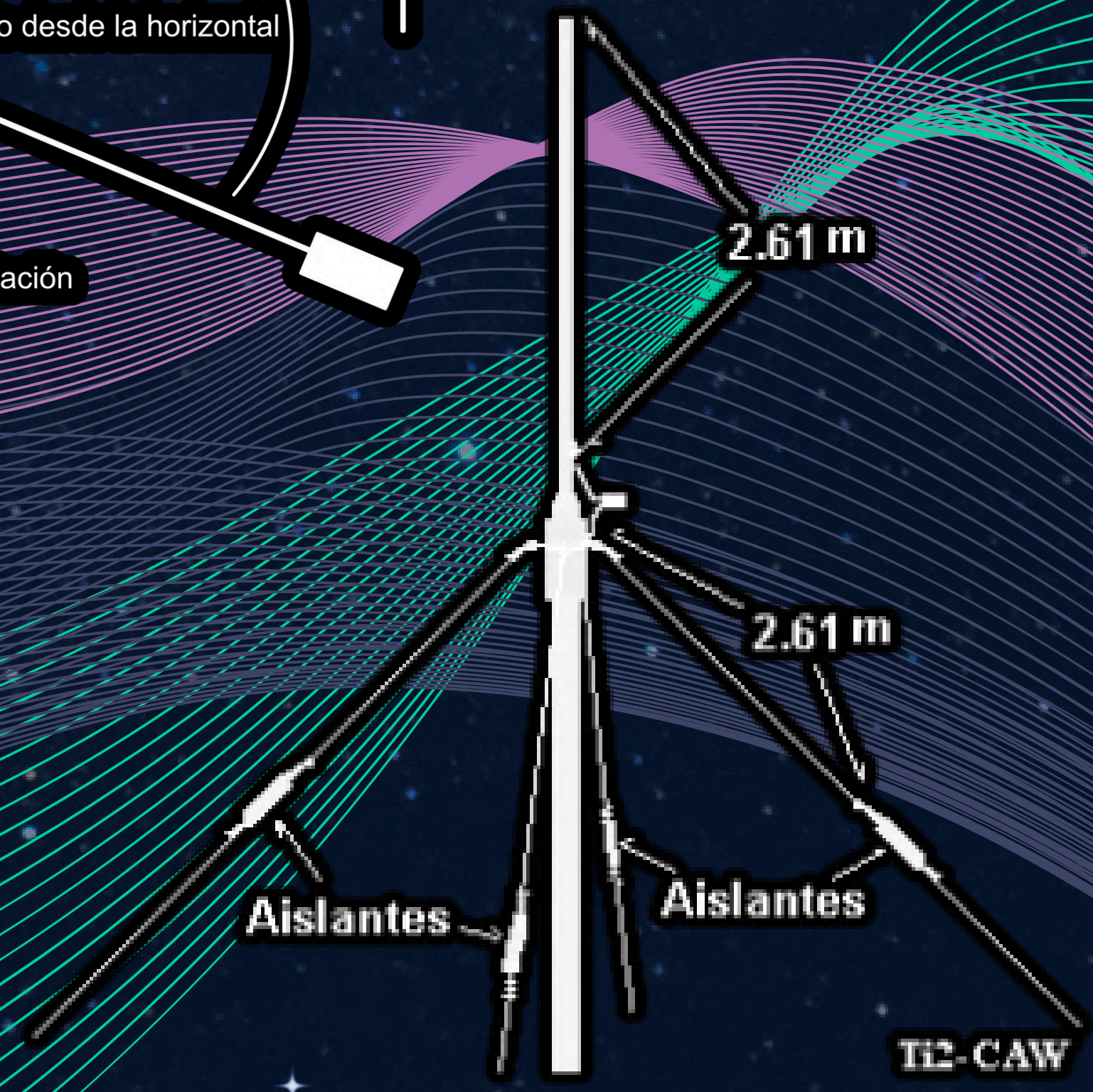
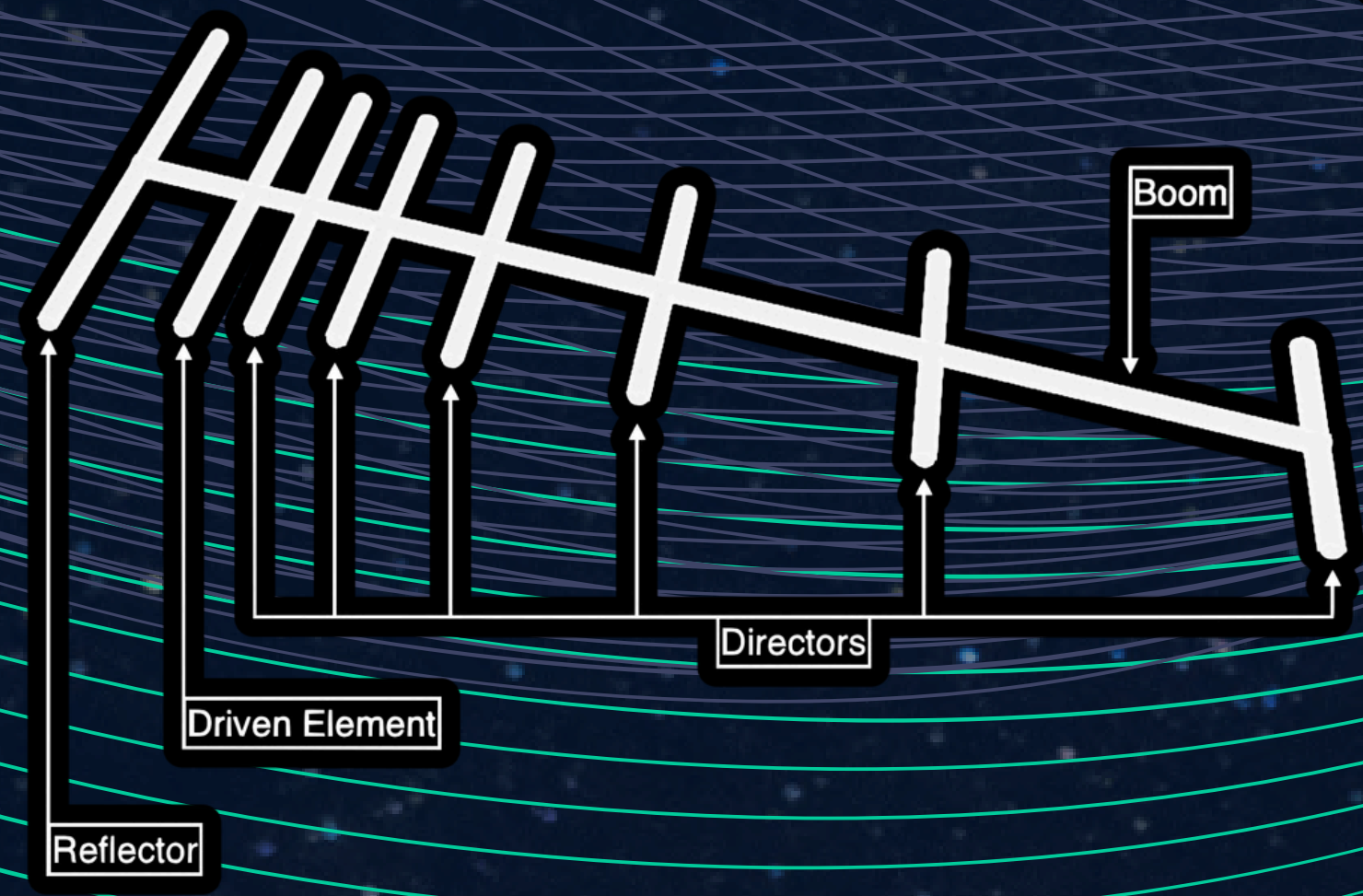
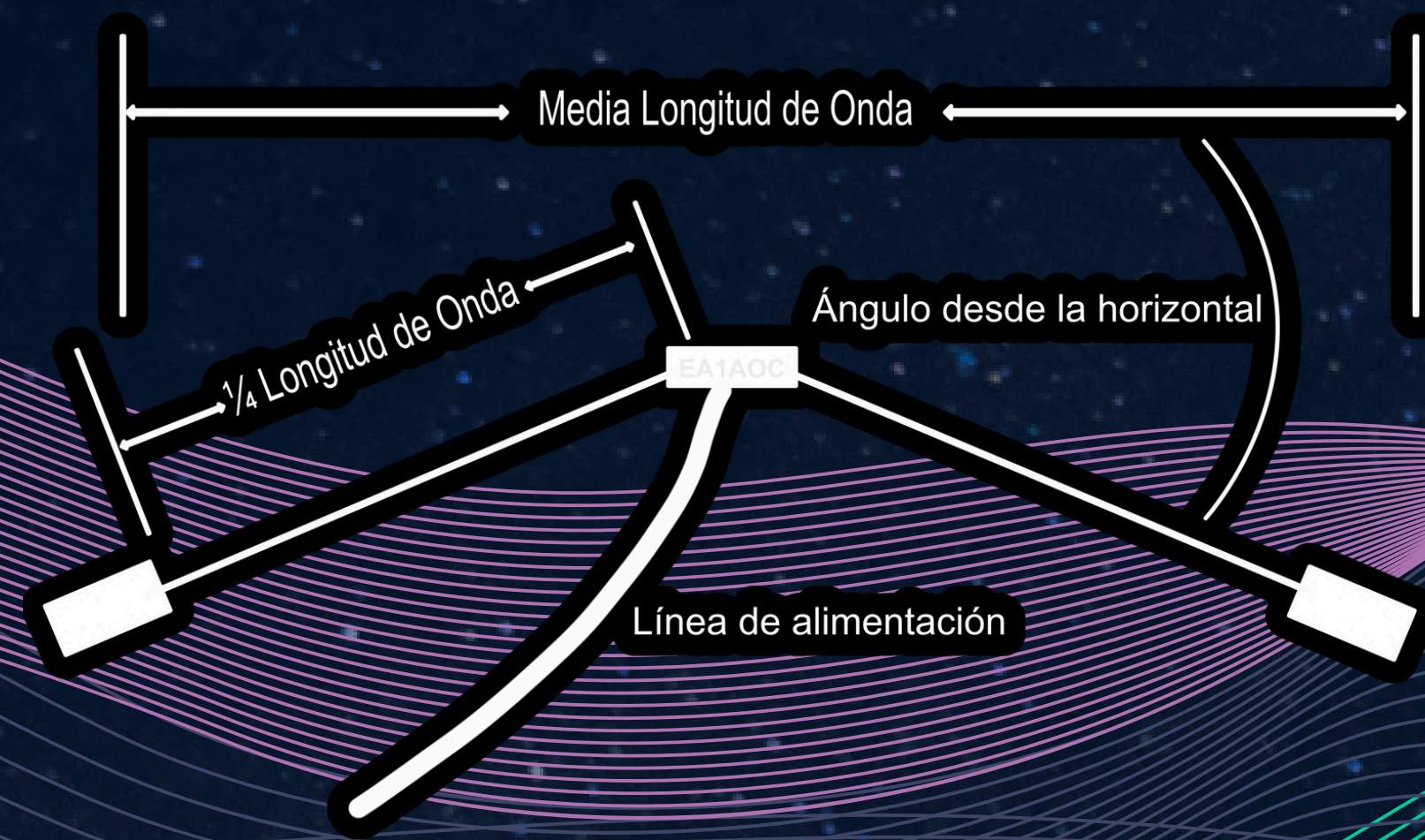
Figure III-4 Polar Orbit Ground Track for 24 Hours (14 orbits)

Modulación de la **señal APT**



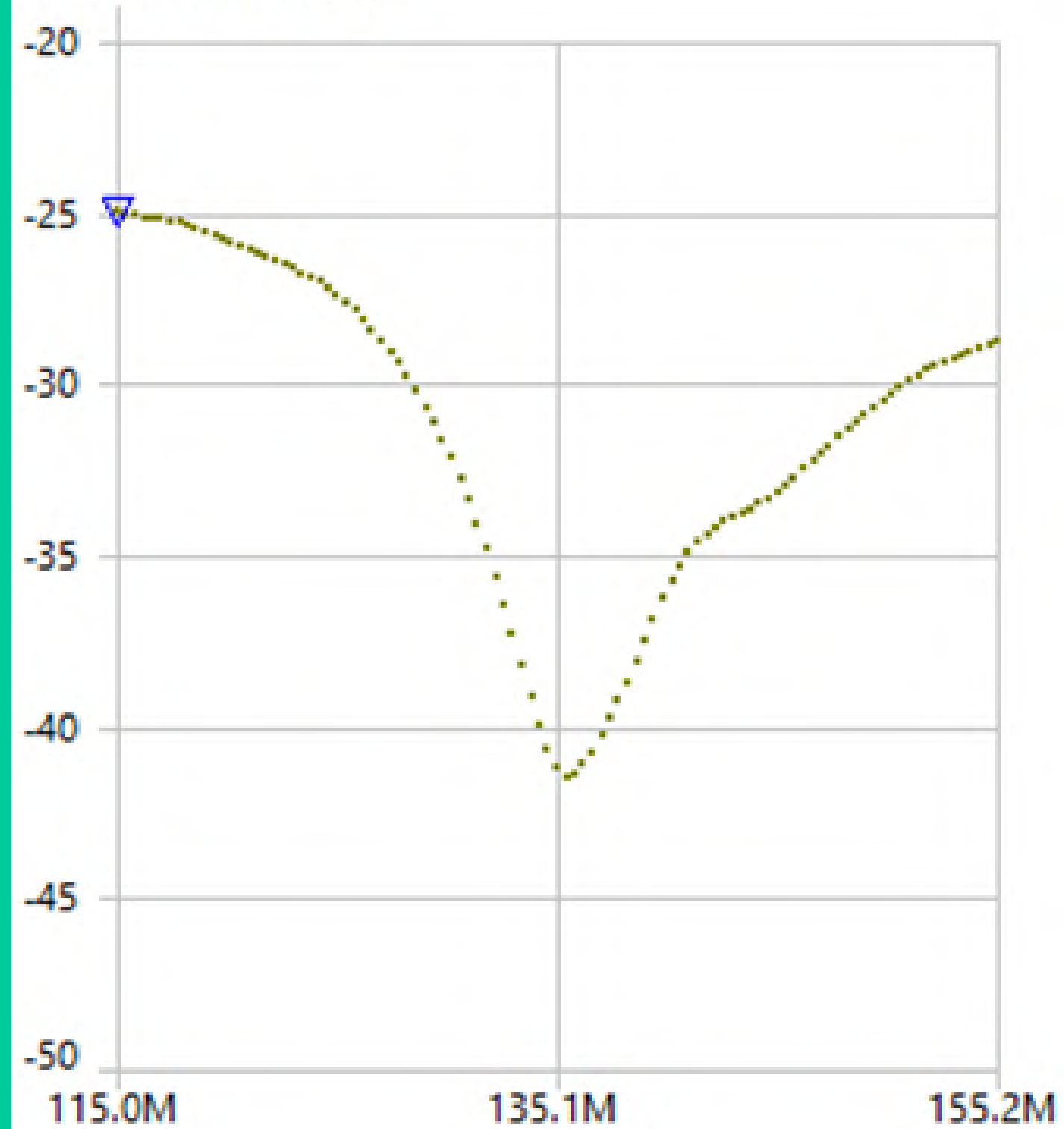
Sobre APT
Transmisión
automática de imágenes

Antenas



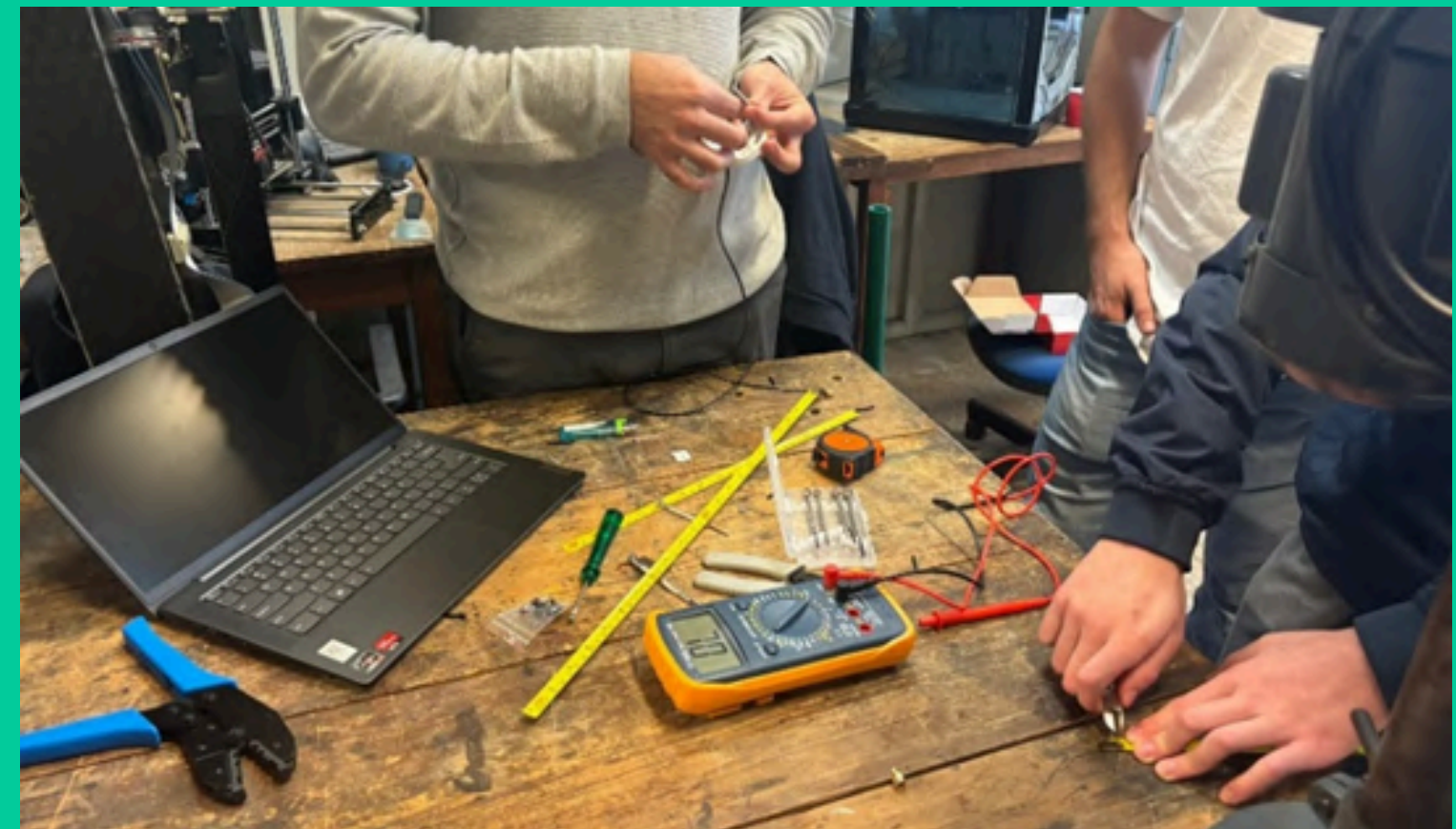
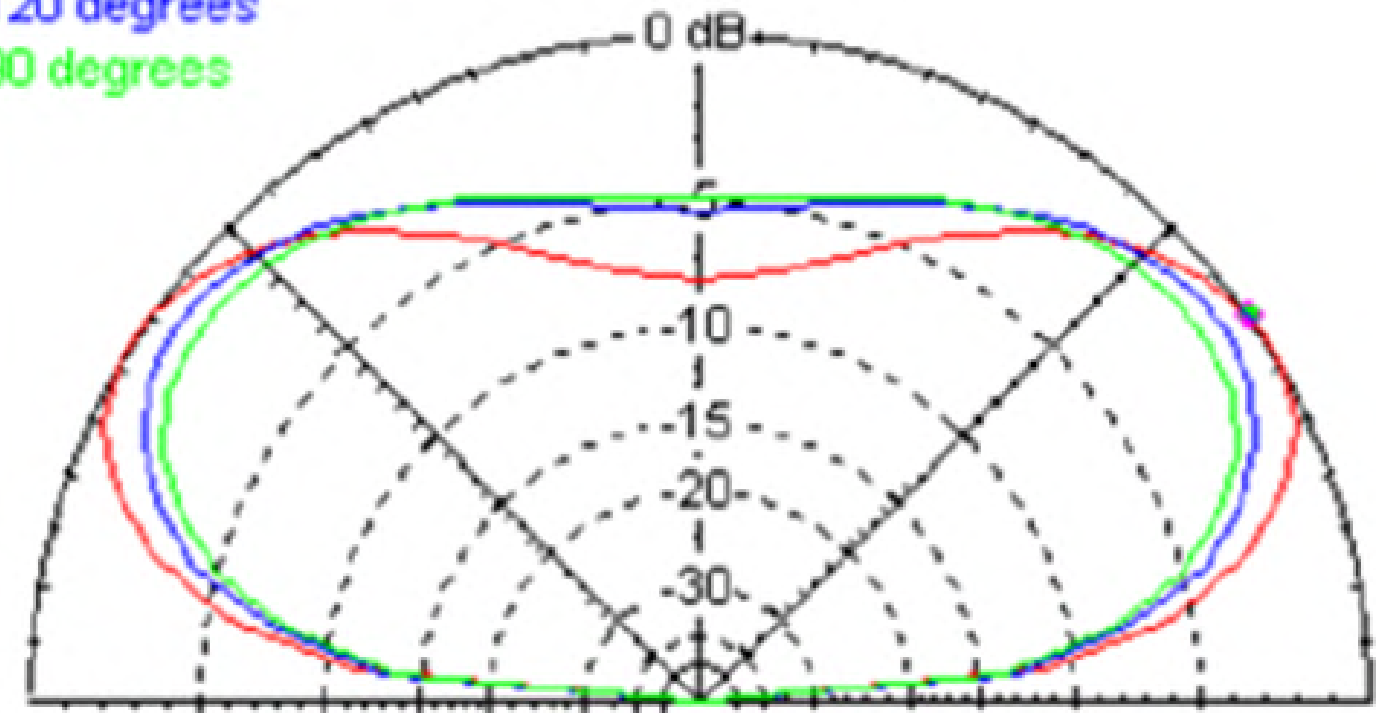
Nuestra Antena

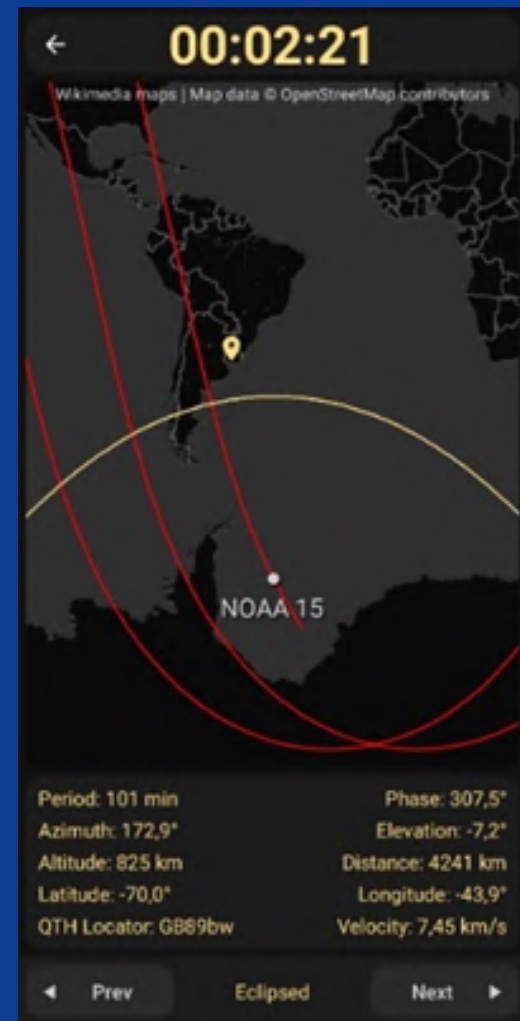
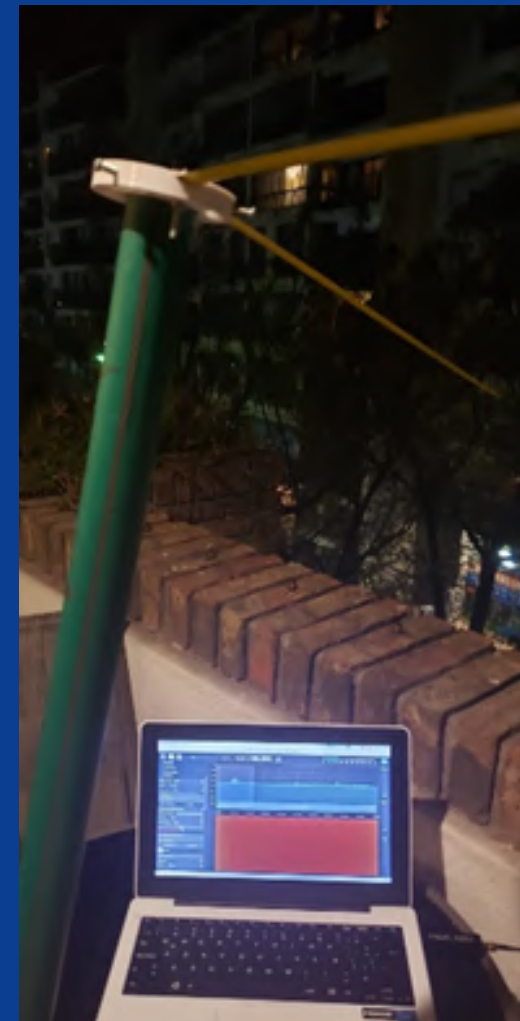
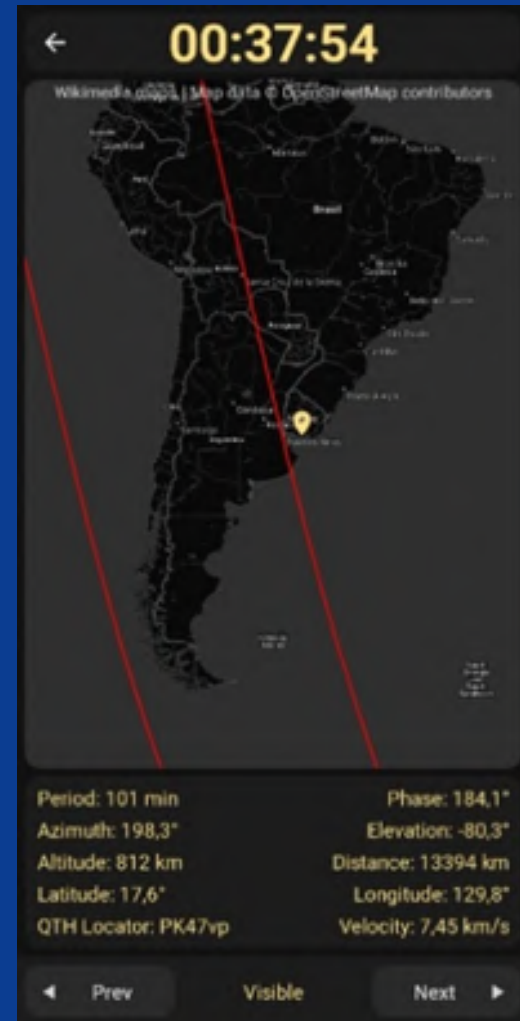
S11 Return Loss (dB)



Total Field

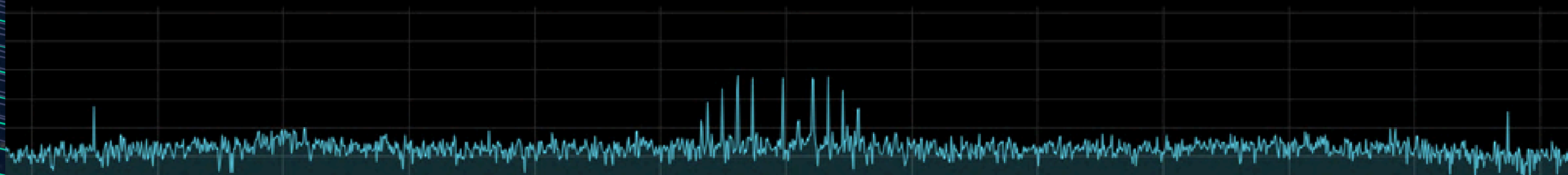
^ Primary
120 degrees
90 degrees





000.137.100.000 ⇌

0 10 20 30 40 50 60 70 80 90



SDR++

Polar Orbiting Satellites

136.98M 137M 137.02M 137.04M 137.06M 137.08M 137.1M 137.12M 137.14M 137.16M 137.18M 137.2M 137.22M

Ancho de **banda**

FM

**(2~10)
AM**

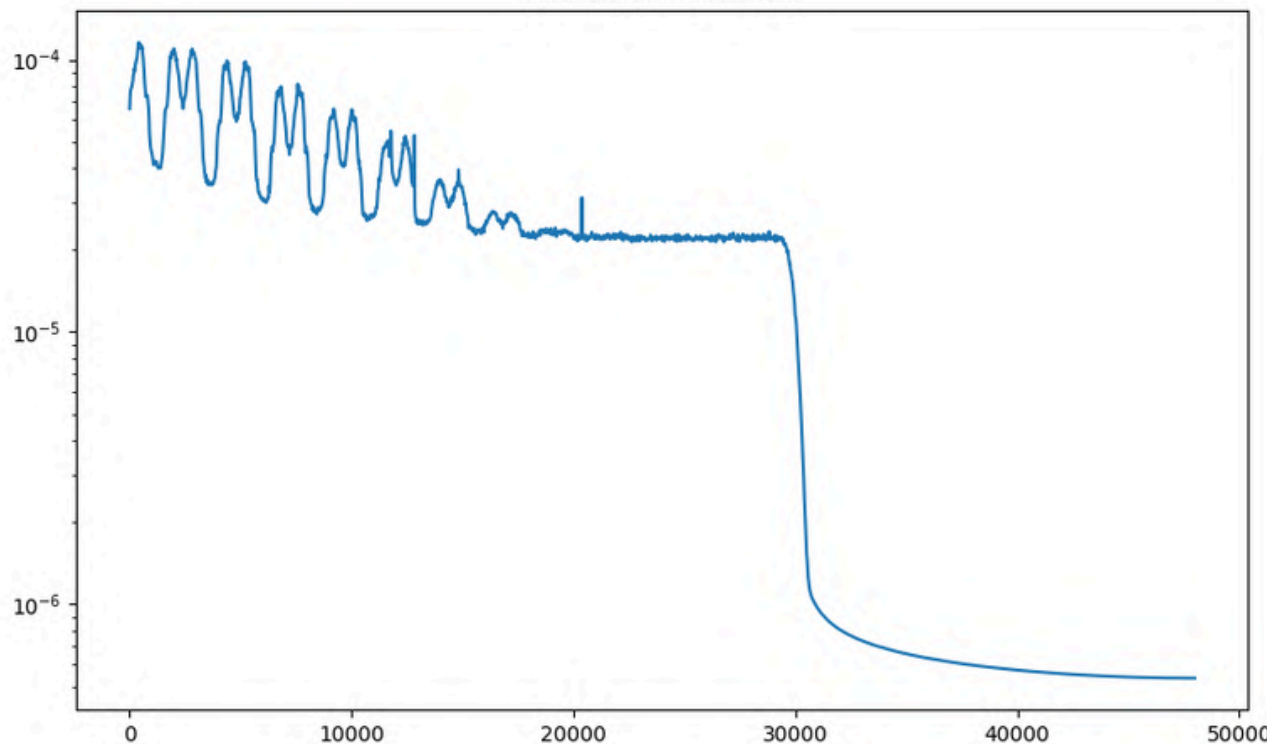
AM

**4480
Hz**

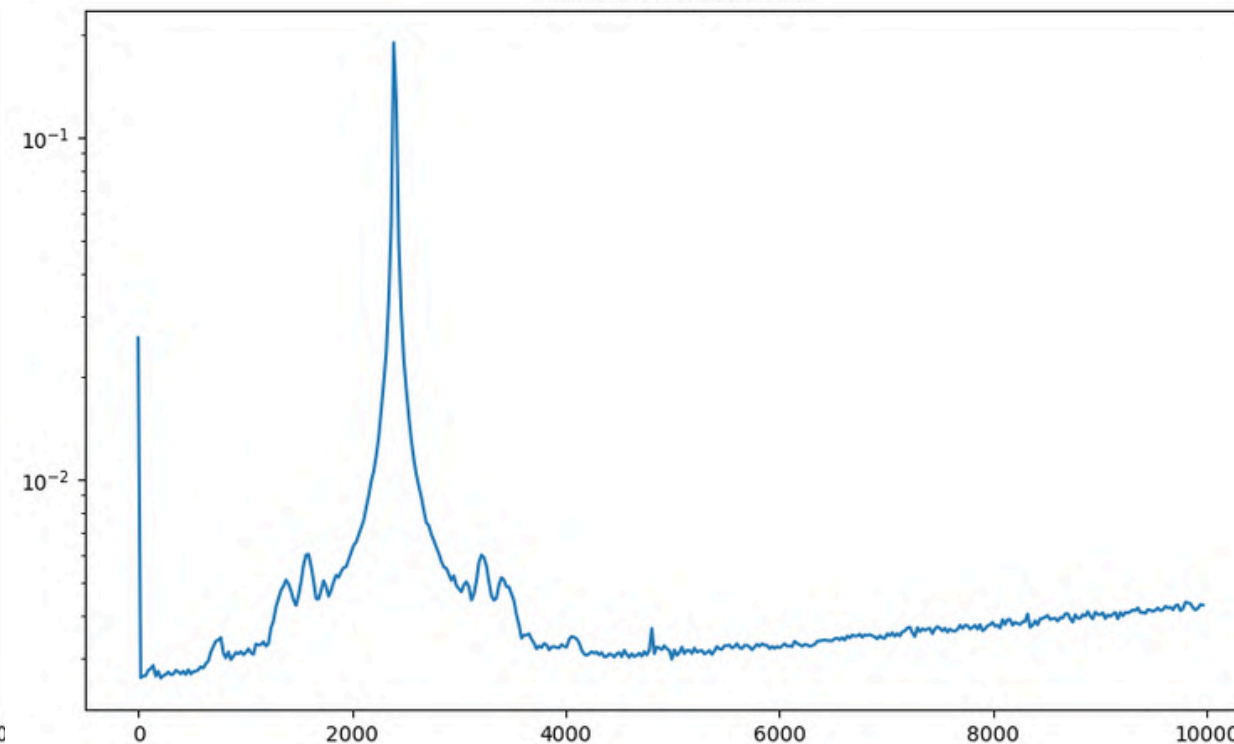
APT

**2080
Hz**

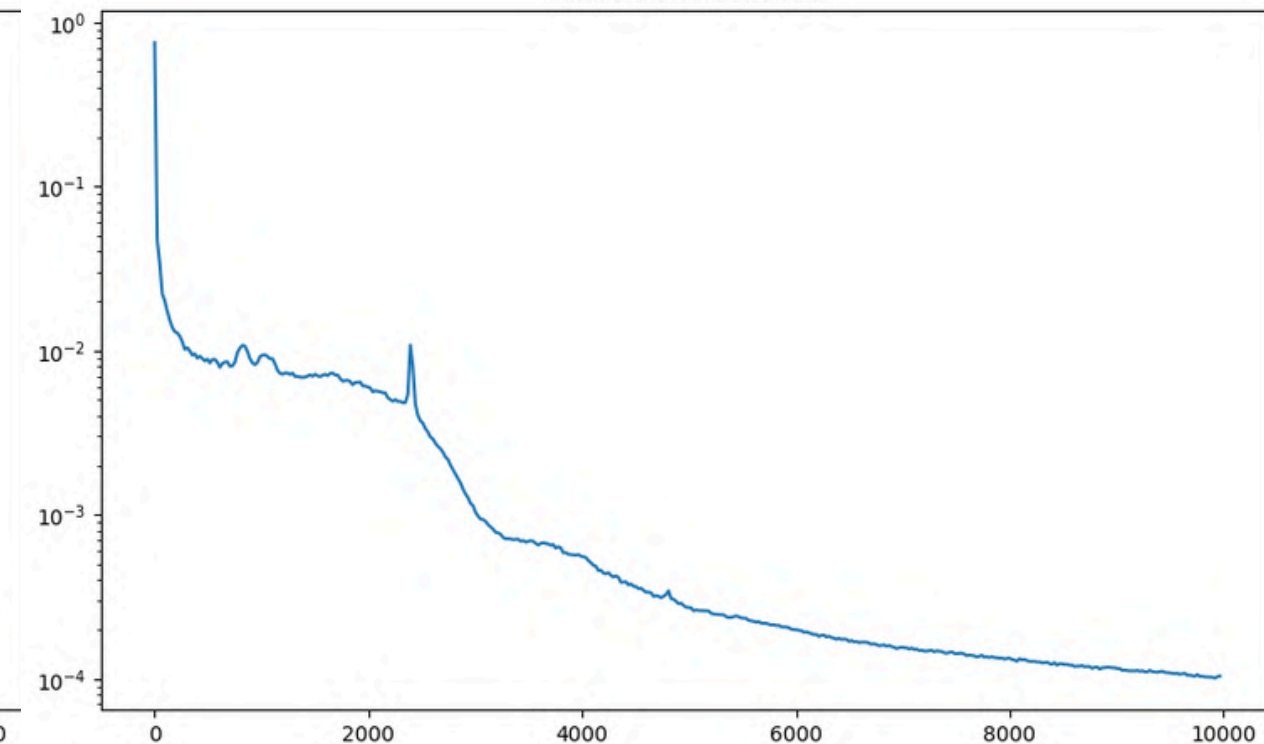
Análisis en frecuencia



Análisis en frecuencia

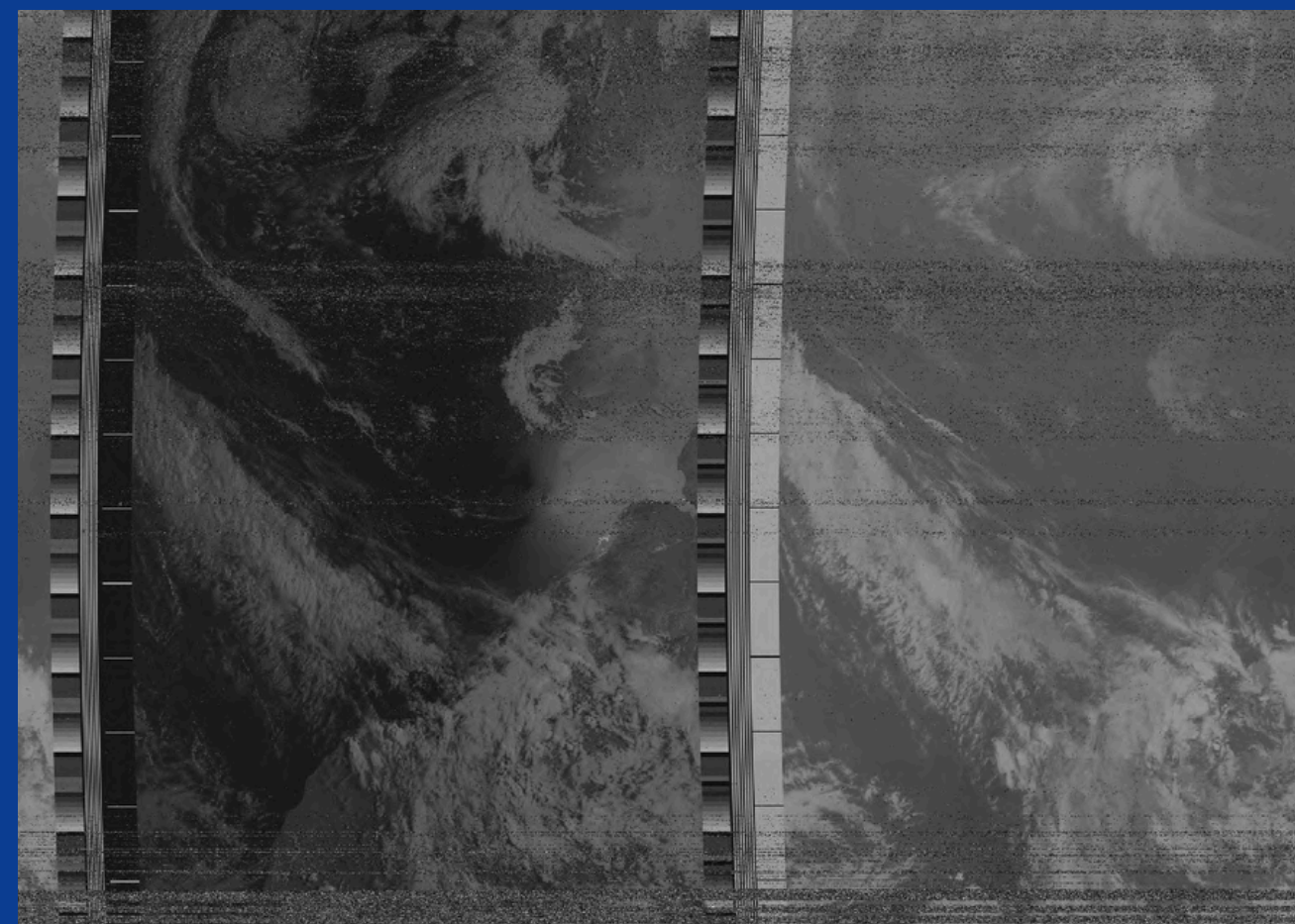
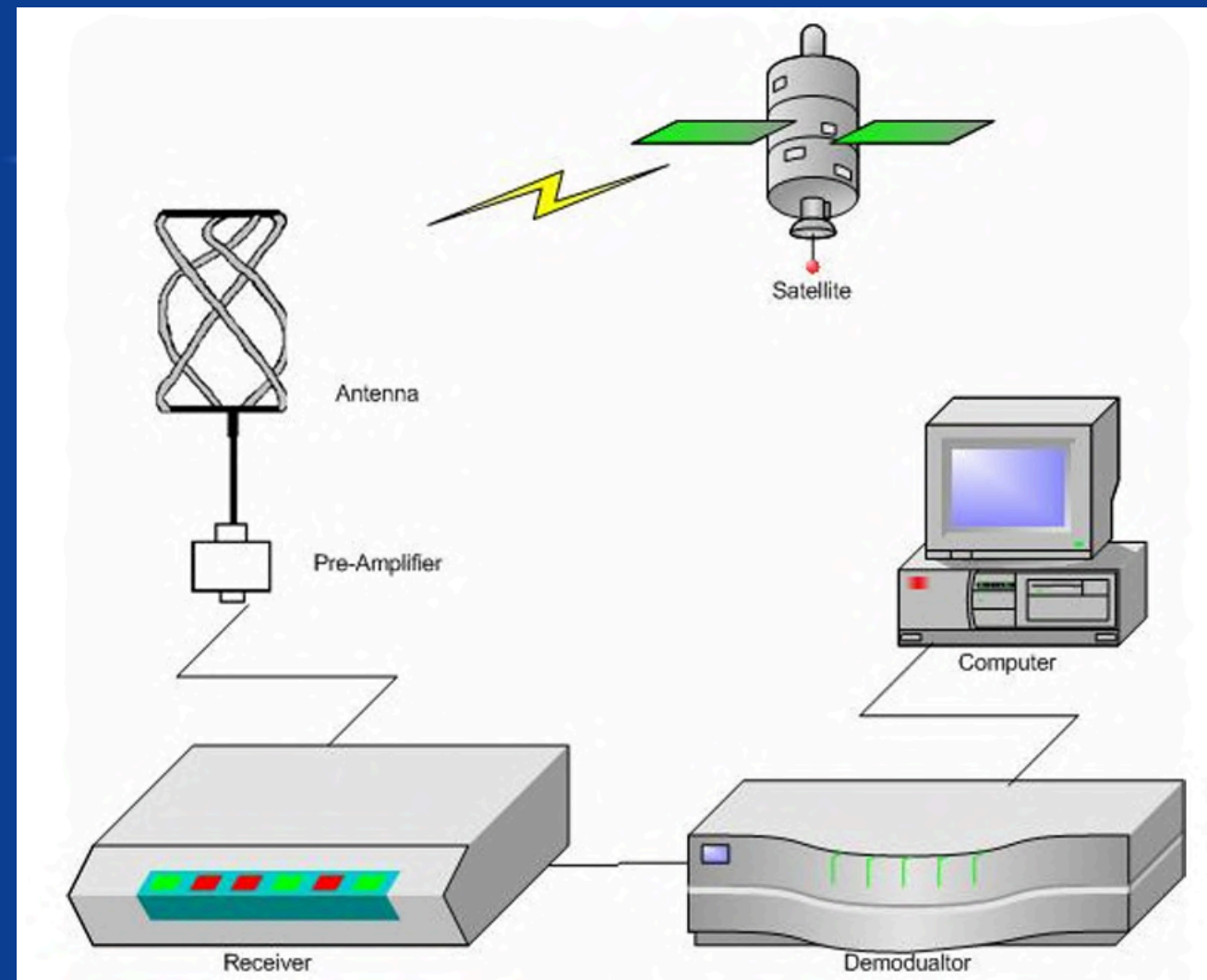


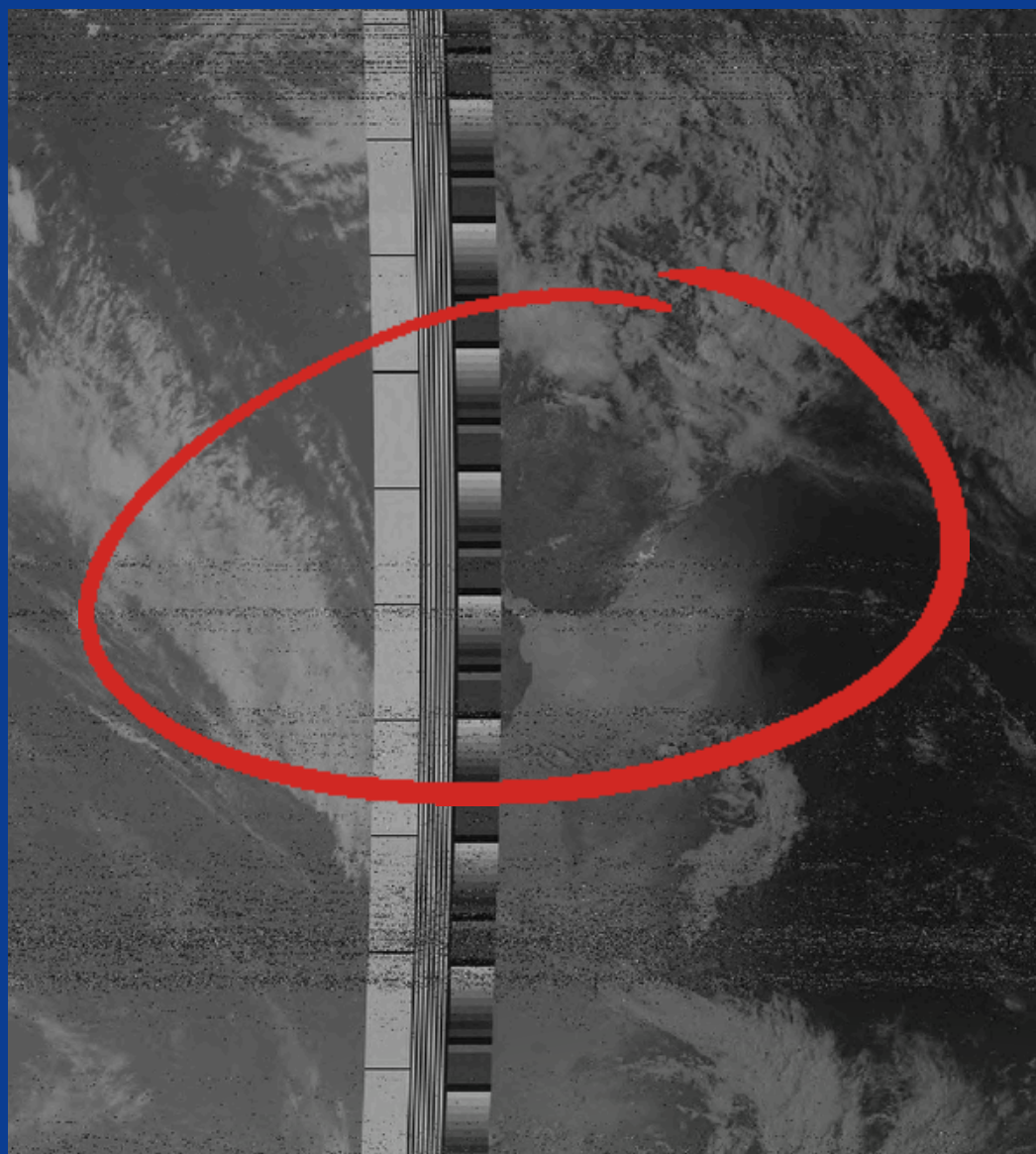
Análisis en frecuencia



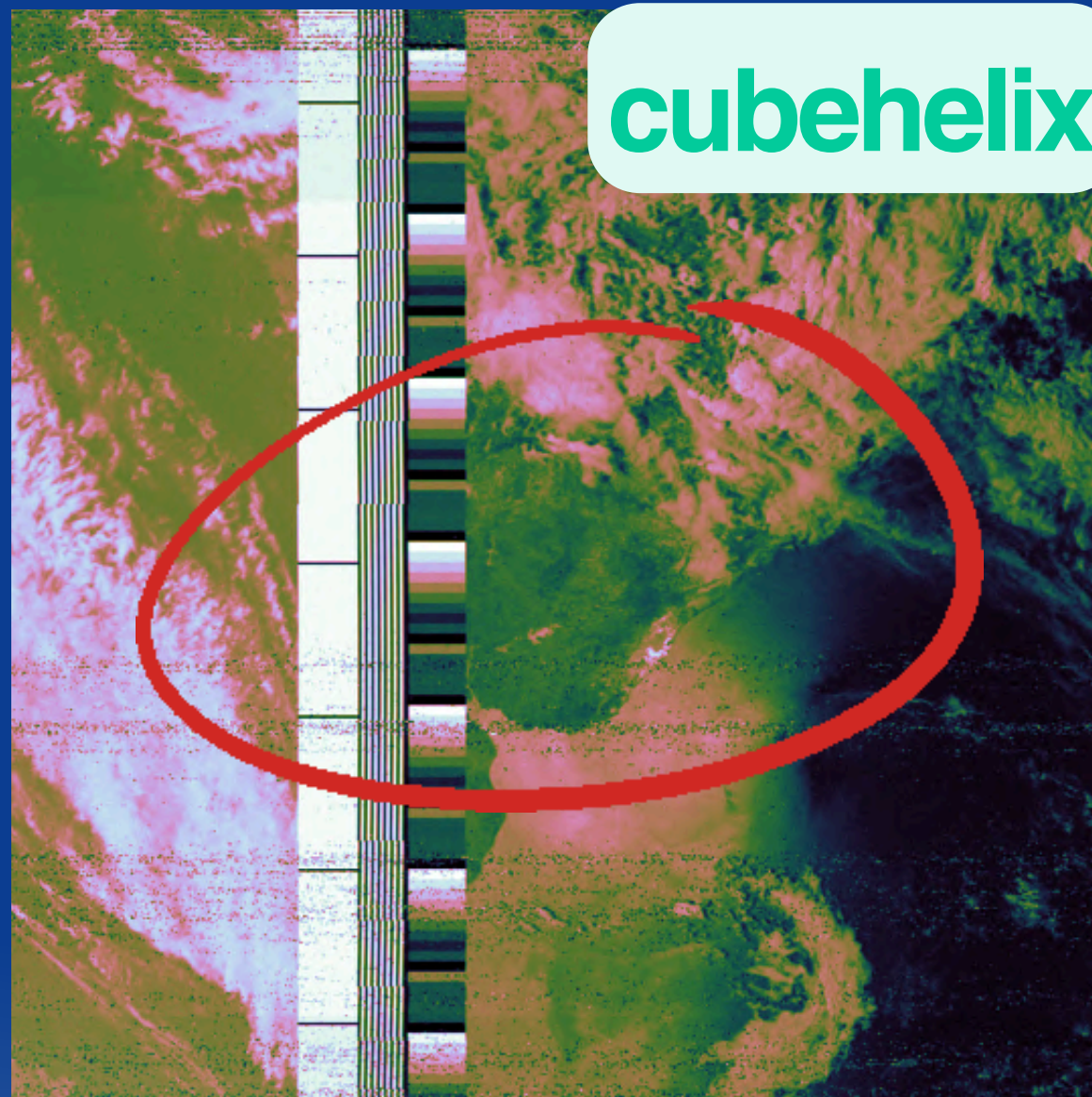
Procesamiento de la señal

- Demodulación FM
- Demodulación AM
- Resample
- Reshape
- Sincronización
- Calibración

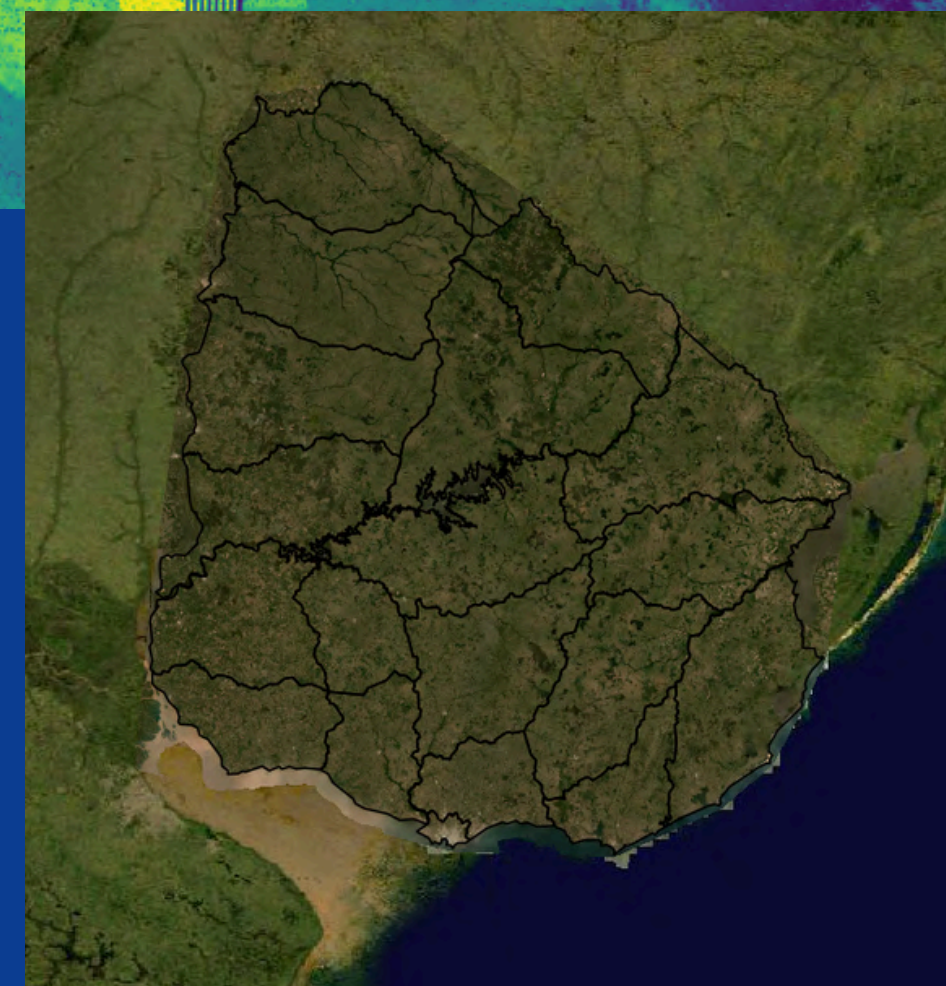
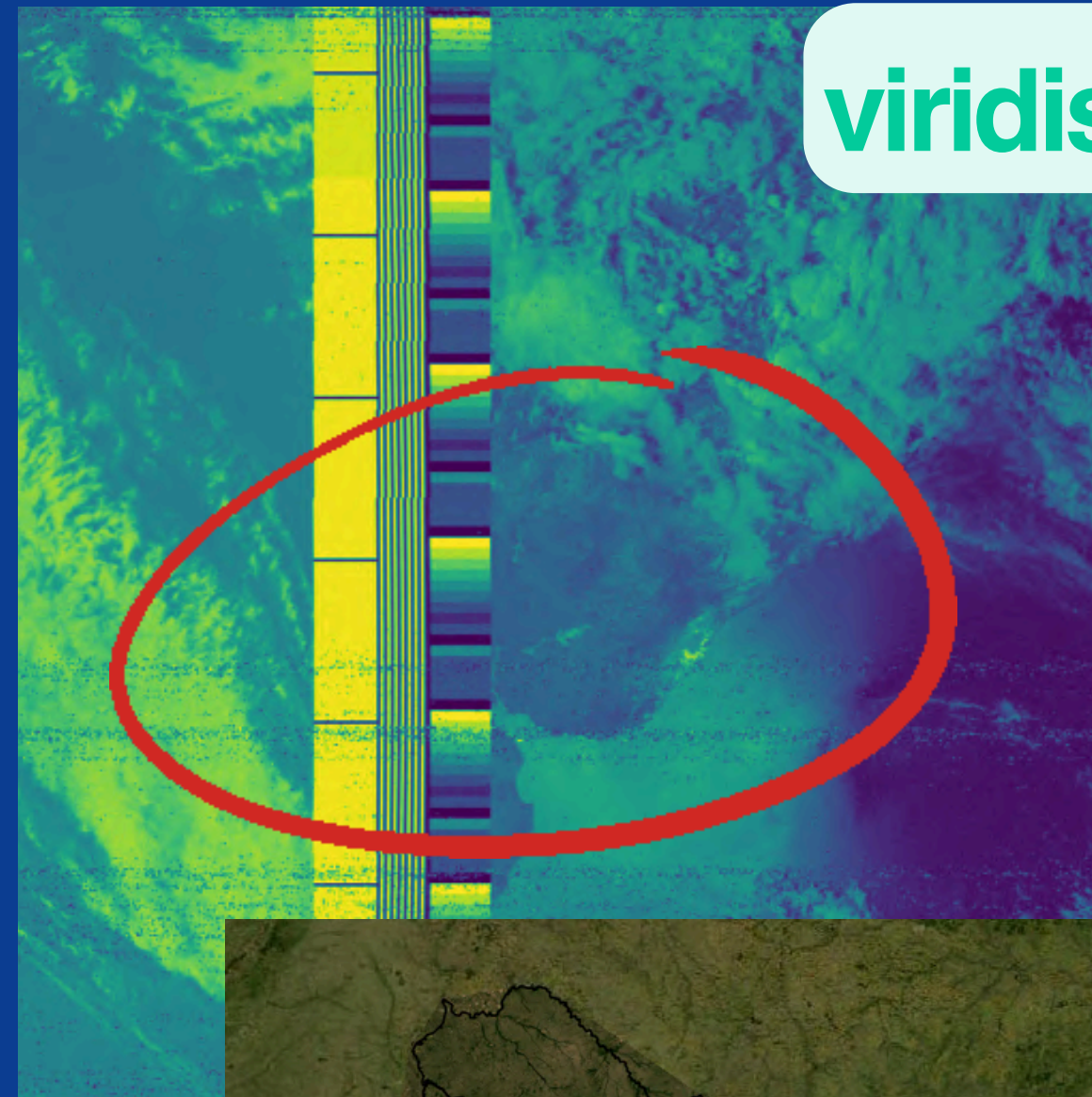


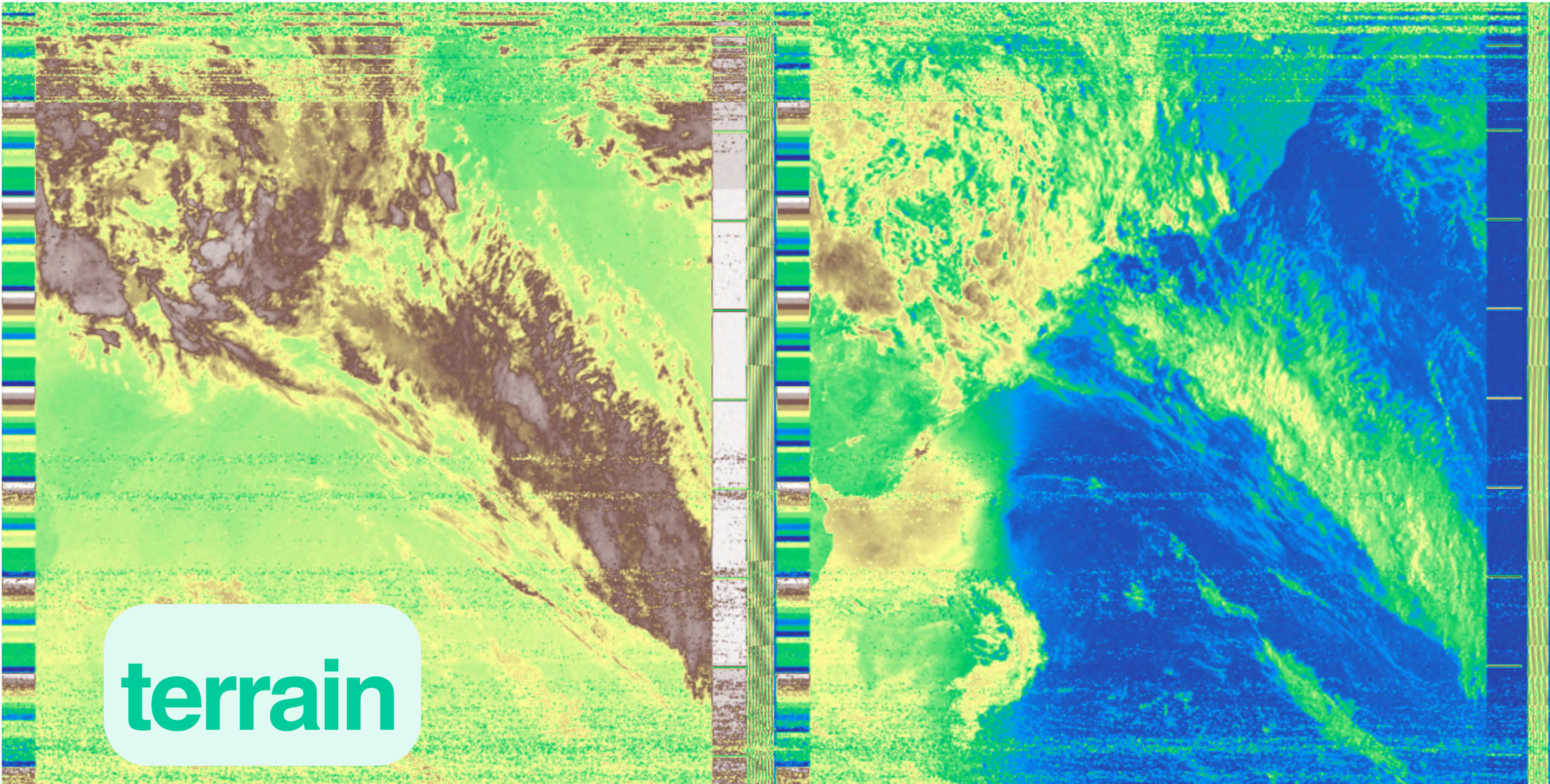


cubehelix

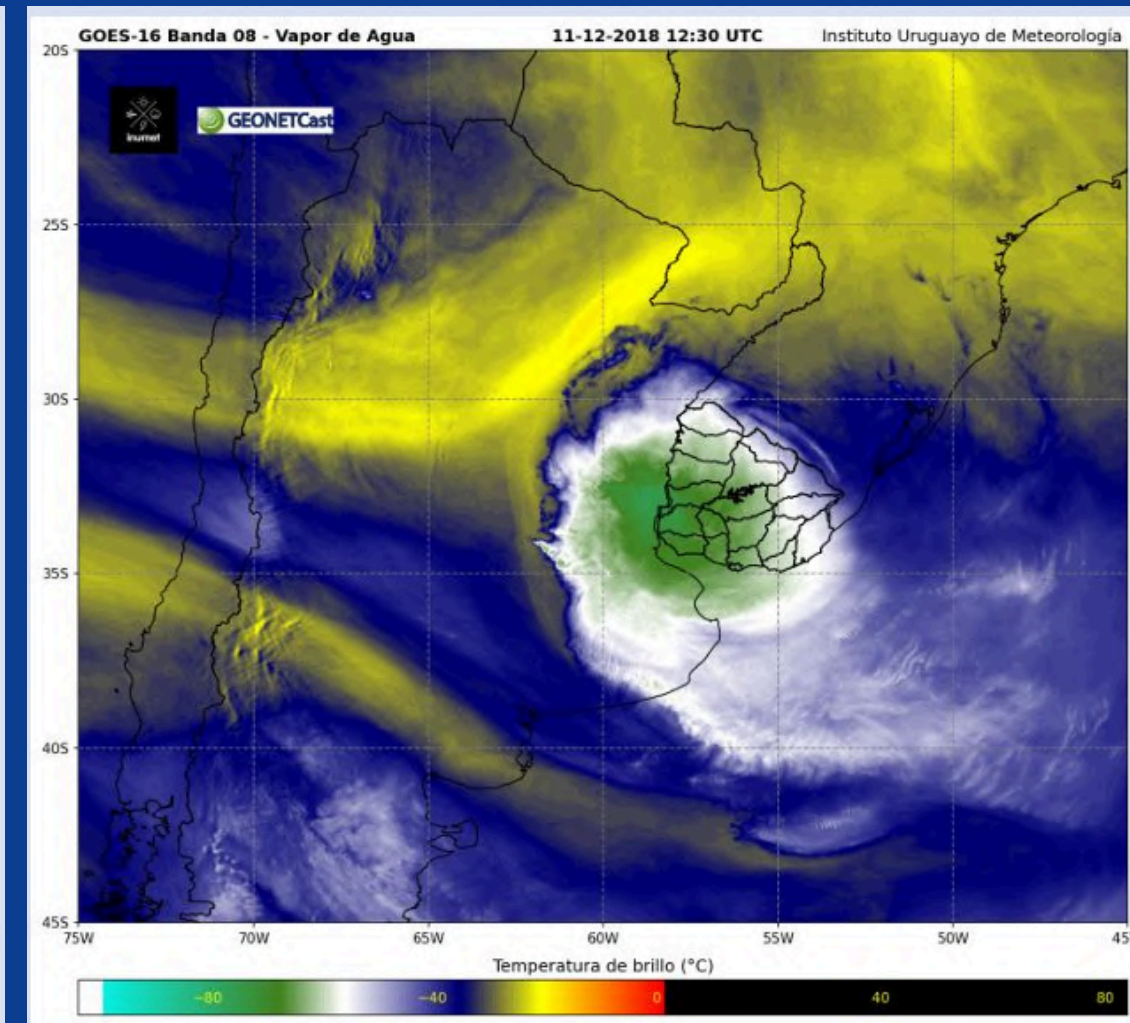
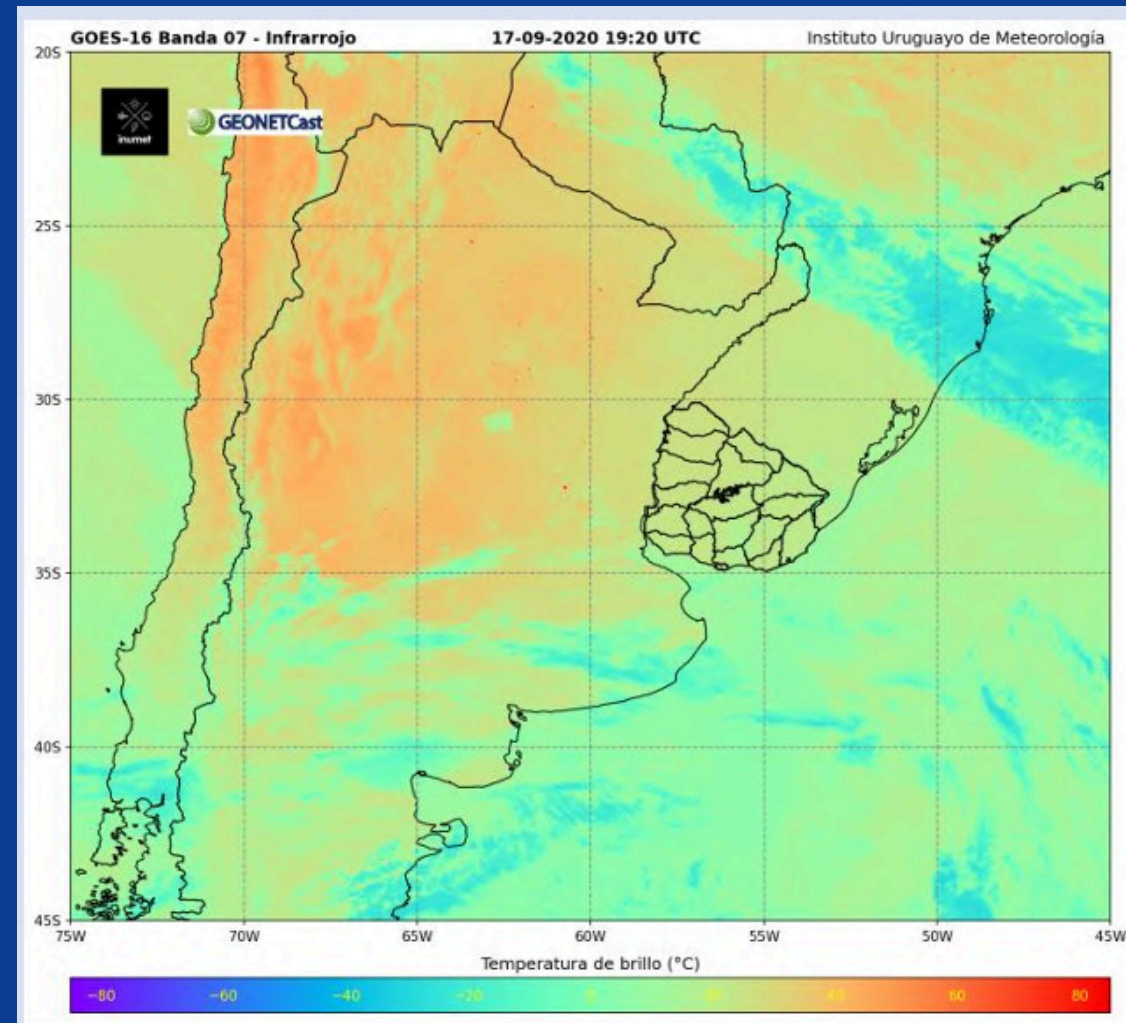
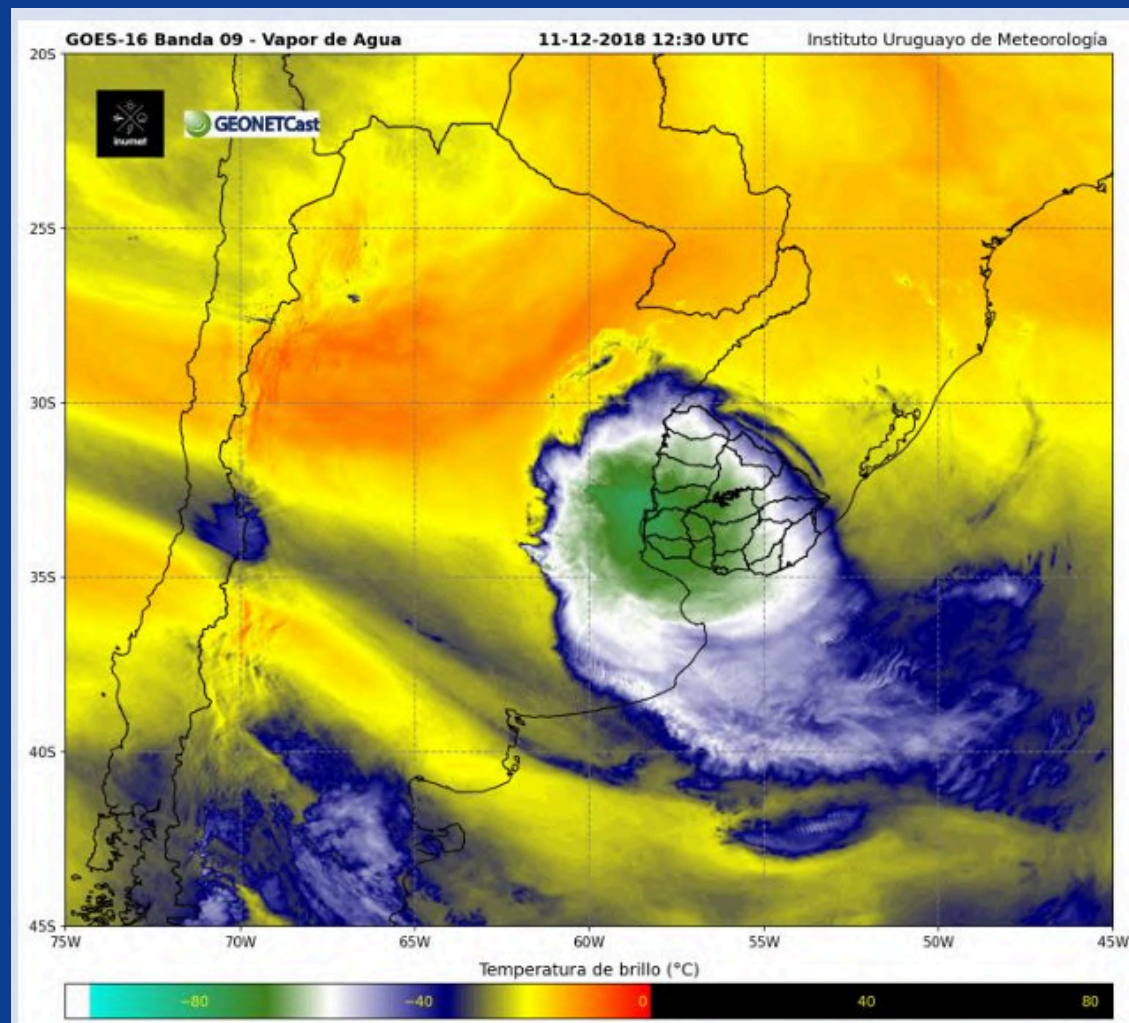


viridis

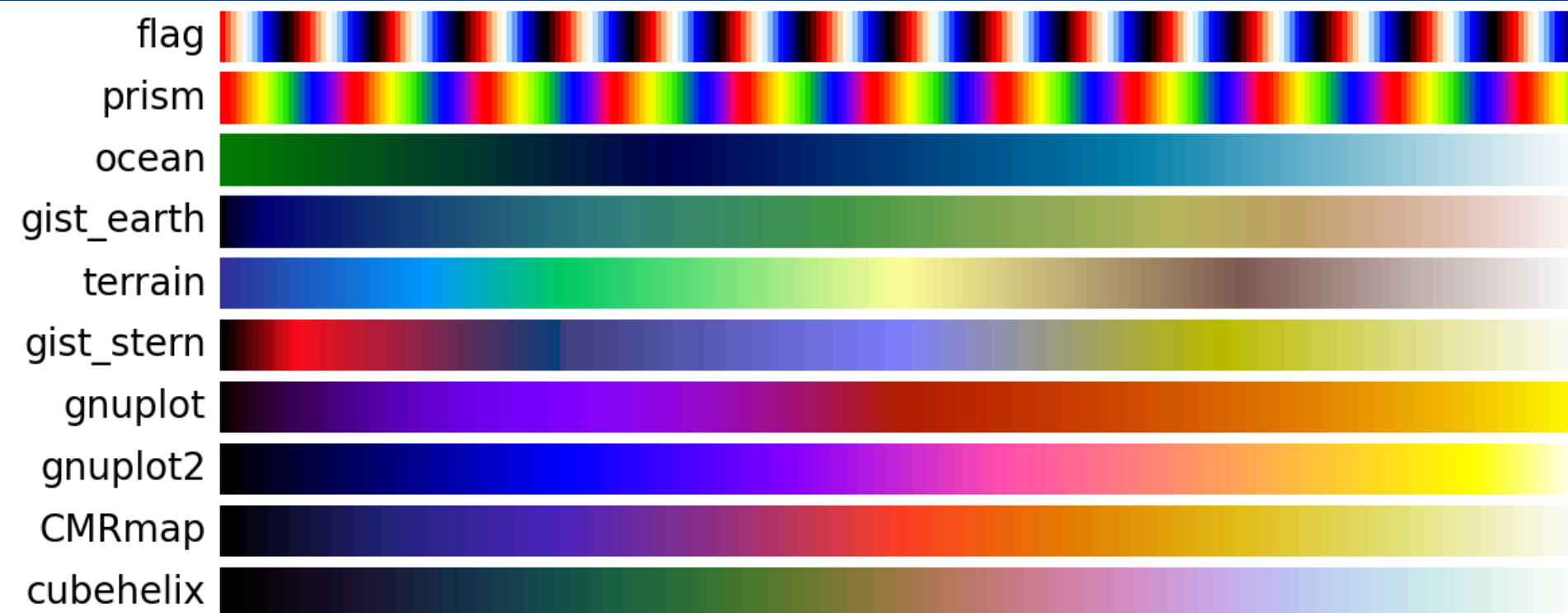




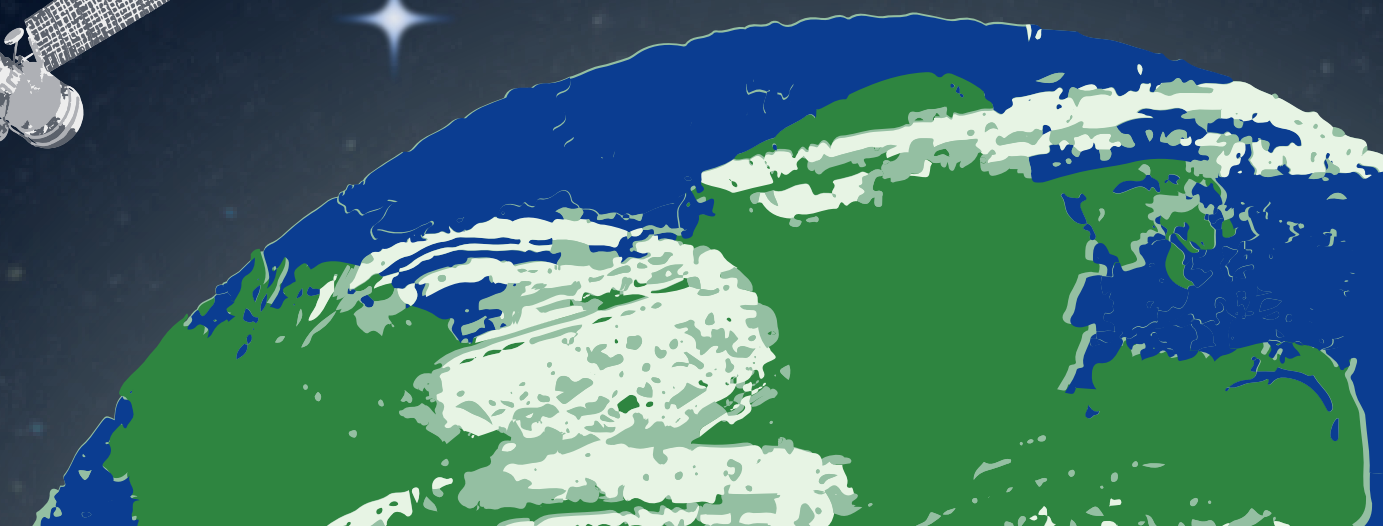
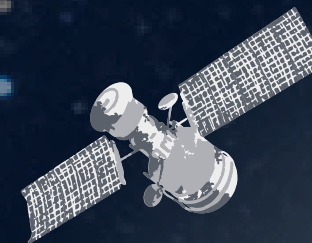
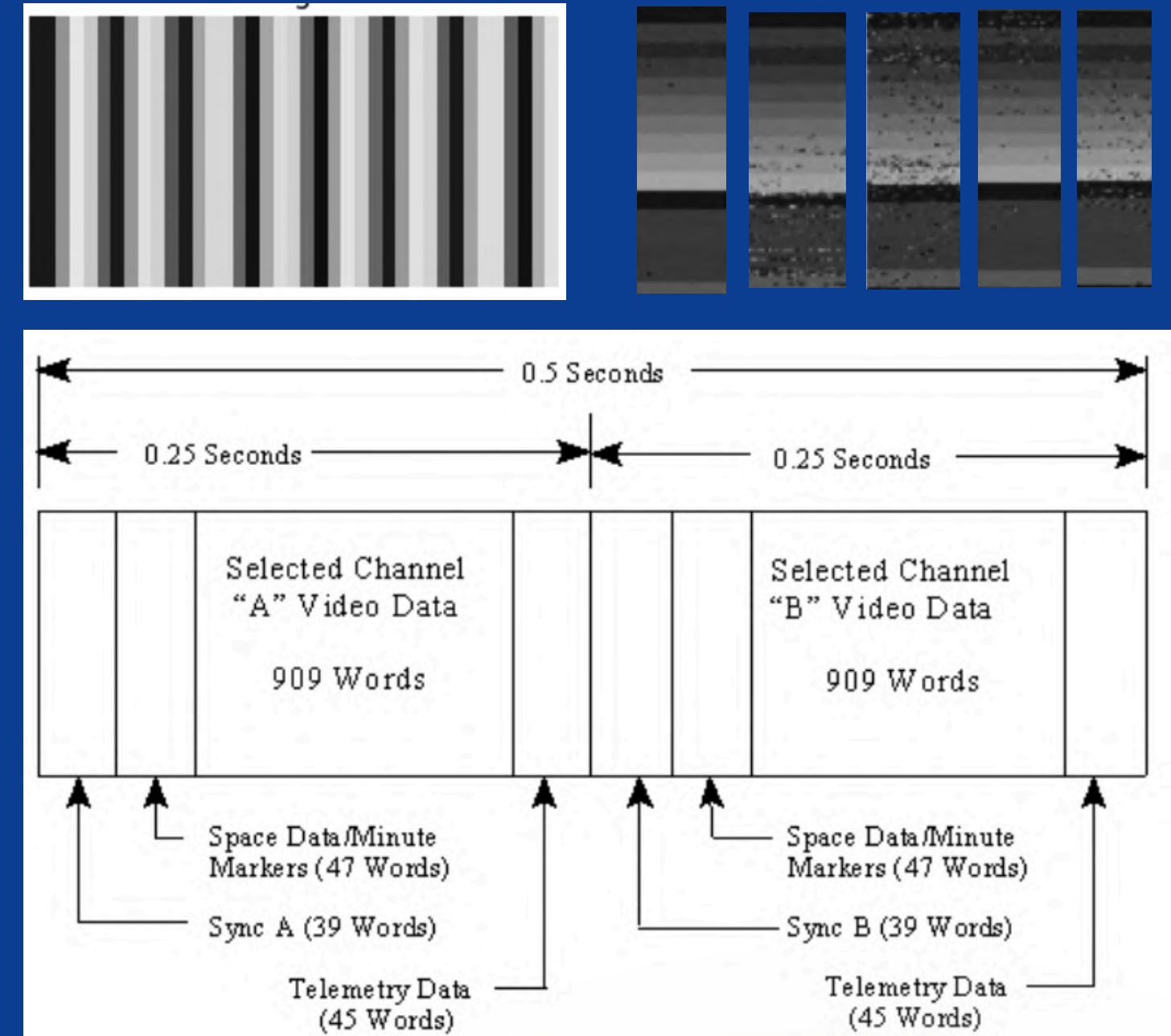
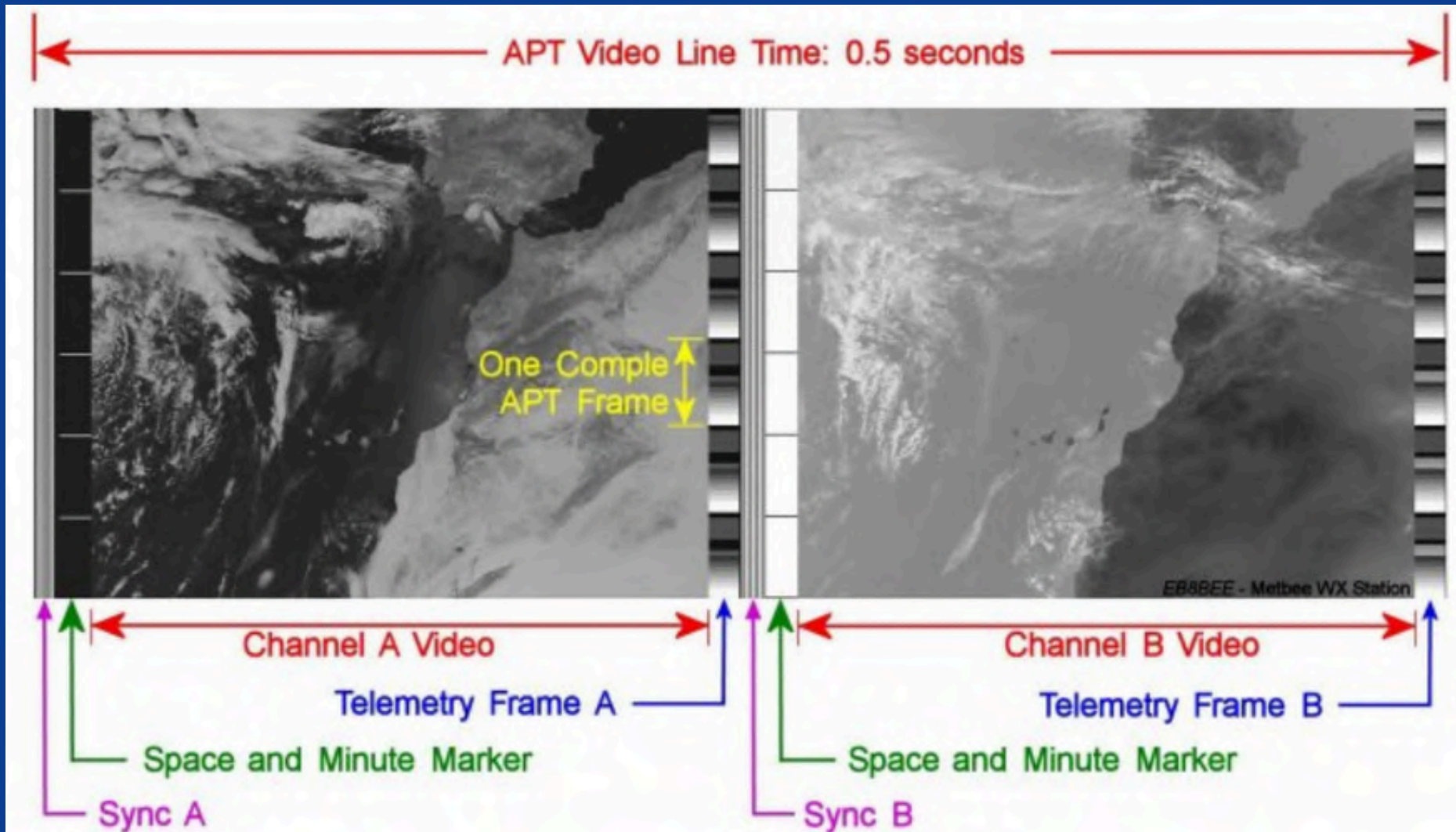
terrain



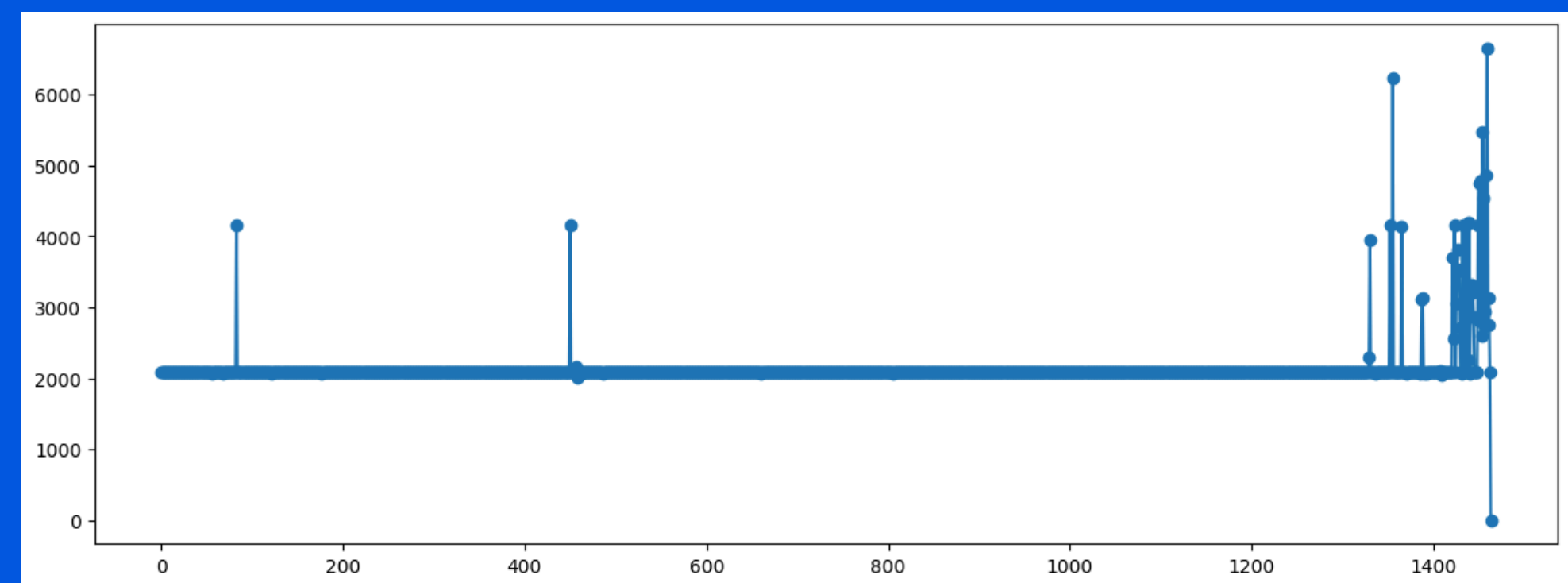
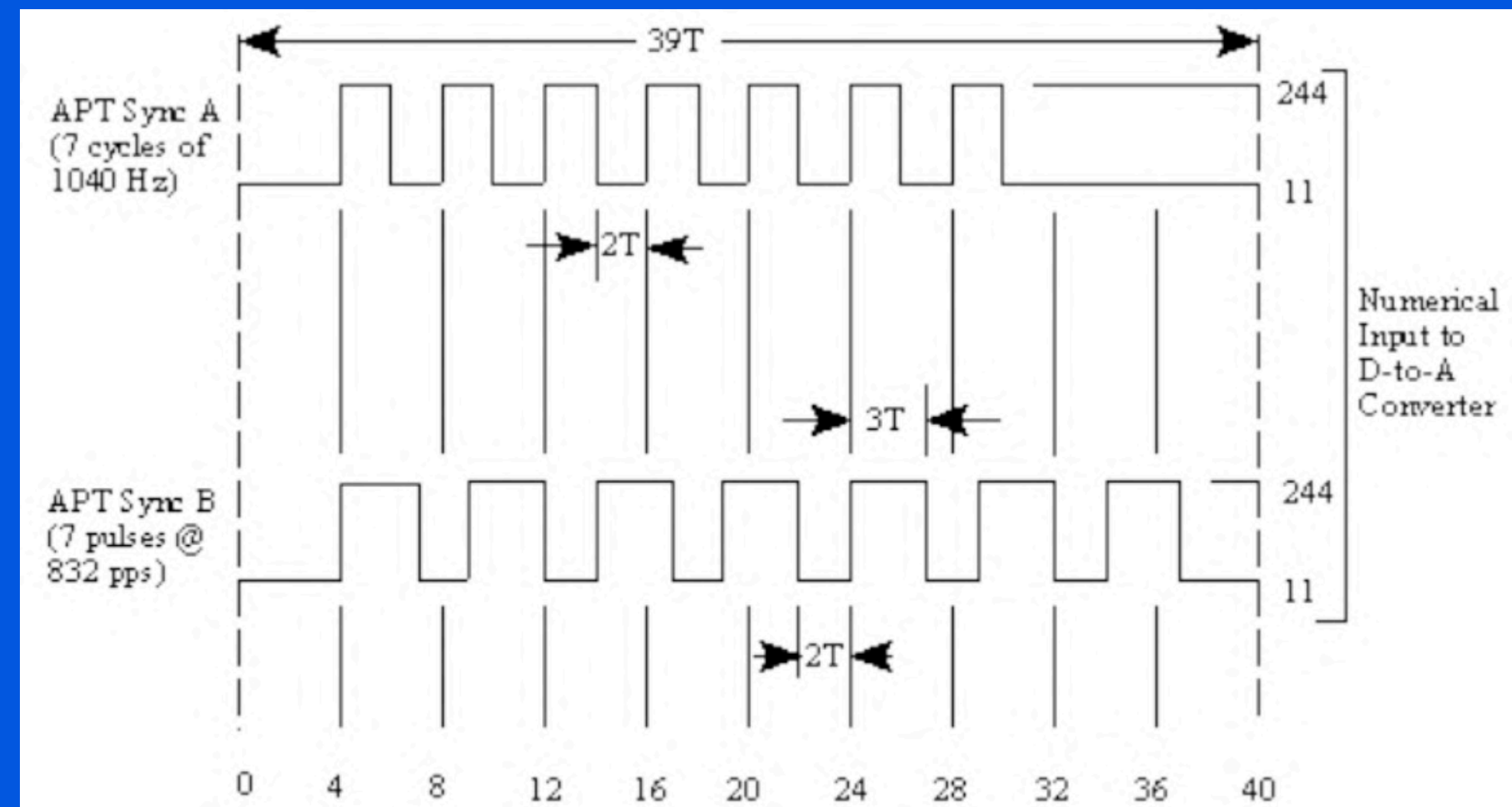
GOES-16



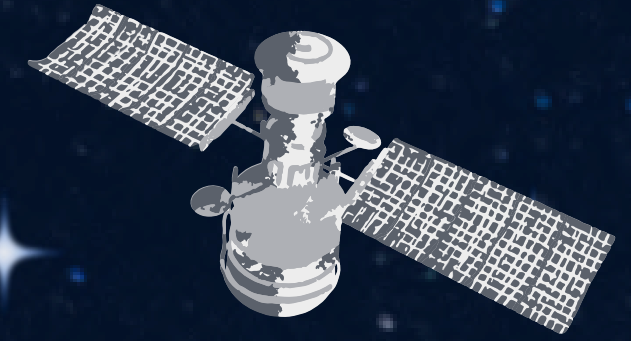
Partes de una línea



APT ANALOG VOLTAGE	DIGITAL VALUE	
1. 0.757 V MI=10.6%	31	
2. 1.538 V MI=21.5%	63	
3. 2.319 V MI=32.4%	95	
4. 3.101 V MI=43.4%	127	
5. 3.881 V MI=54.2%	159	
6. 4.663V MI=65.2%	191	
7. 5.444 V MI=76.0%	223	
8. 6.225V MI=76.0%	255	
9. ZERO MODULATION	0	
10. THERM TEMP PRT #1		
11. THERM TEMP PRT #2		
12. THERM TEMP PRT #3		
13. THERM TEMP PRT #4		
14. PATCH TEMP		
15. BACK SCAN		
16. CHANNEL IDENT		



Conclusiones



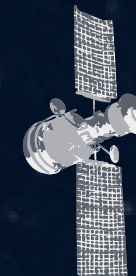
¹ Satélites
NOAA ✓

² Antena y
Grabaciones ✓

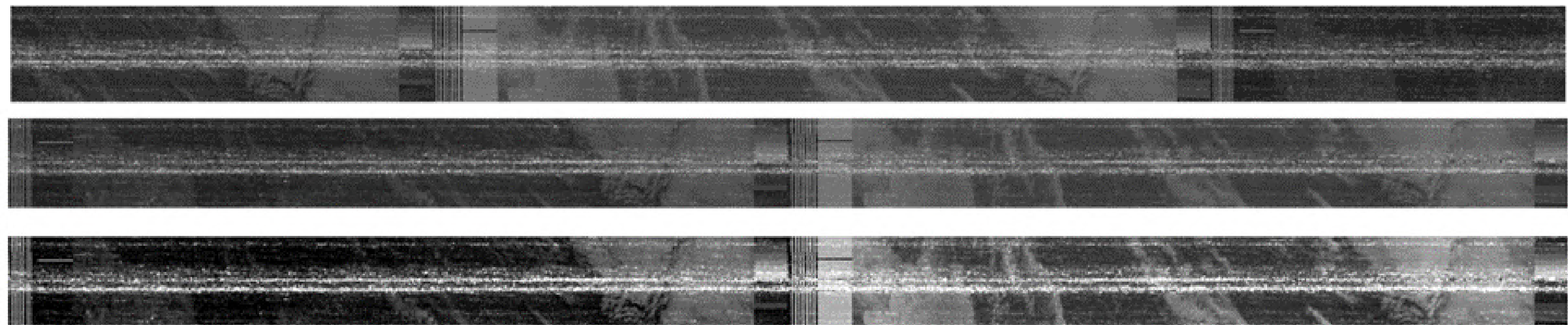
³ Procesamiento
de la **Señal** ✓

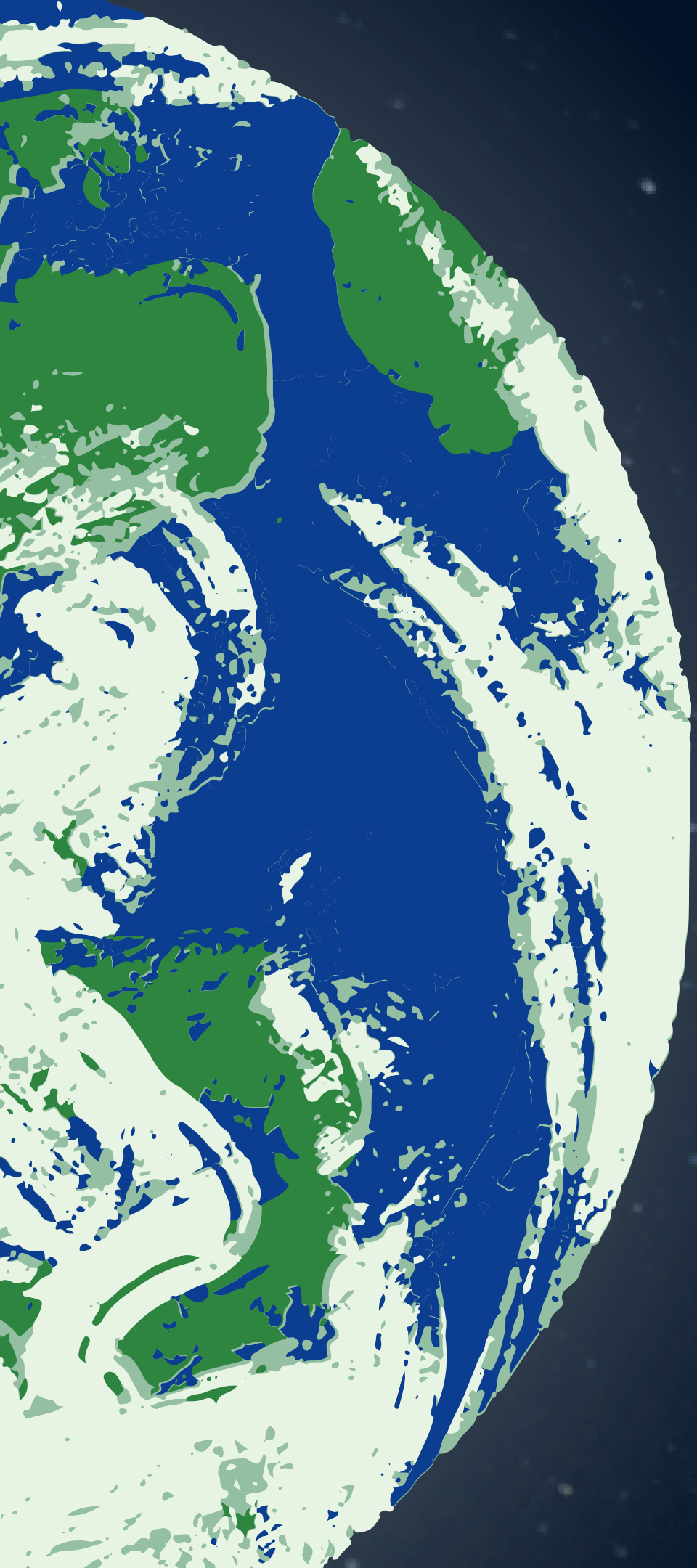
⁴ Nuestras
Imágenes ✗

Lastimosamente...



Aún así





Muchas
Gracias

